COURSE SLIDES · WSQ

AWS Certified SysOps Administrator Associate Training

WSQ Course Code: TGS-2024051413

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v5.0 · 23 August 2026

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ADMI
N

COURSE ADMINISTRATION

Welcome & Course Orientation

Three days · 24 hours · 12 labs · 2 assessment hours

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Digital Attendance (Mandatory)

1

AM, PM and Assessment

Complete every required SSG digital attendance and wait for confirmation.

2

Trainer displays the QR

Scan only the current session's QR code using your mobile camera.

3

75% minimum

Attendance and assessment requirements determine course-completion eligibility.

4

Evidence

Notify the trainer immediately if a submission does not confirm.

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About the Trainer



Your Trainer

General trainer profile template

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AWS AND CLOUD EXPERTISE

INDUSTRY EXPERIENCE

CONTACT

personal information is optional

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Founder & Chief Training Officer
Tertiary Infotech Academy Pte Ltd

EXPERIENCE

More than 20 years across software engineering, AI, data analytics and cloud computing.

QUALIFICATIONS

PhD (NUS), DACE, ACTA and GCWLP among professional credentials.

COURSE FOCUS

AWS observability, automation, resilience, security and evidence-led operations.

TEACHING APPROACH

Connects architecture choices to measurable service outcomes and controlled change.

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ICE-BREAKER

Let's Know Each Other

1

Introduce

Share your name, organisation and role.

2

Experience

Name one cloud service you have operated or supported.

3

Aim

Choose one outcome: stronger monitoring, recovery, automation, security or troubleshooting.

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SAFE CLOUD LEARNING

Ground Rules

1. **Participate**
   Ask, challenge and contribute evidence.

2. **Respect**
   Critique the system or idea, not the person.

3. **Protect access**
   Never share AWS passwords, access keys or account details.

4. **Bound risk**
   Use learner-owned resources and the trainer-approved Region.

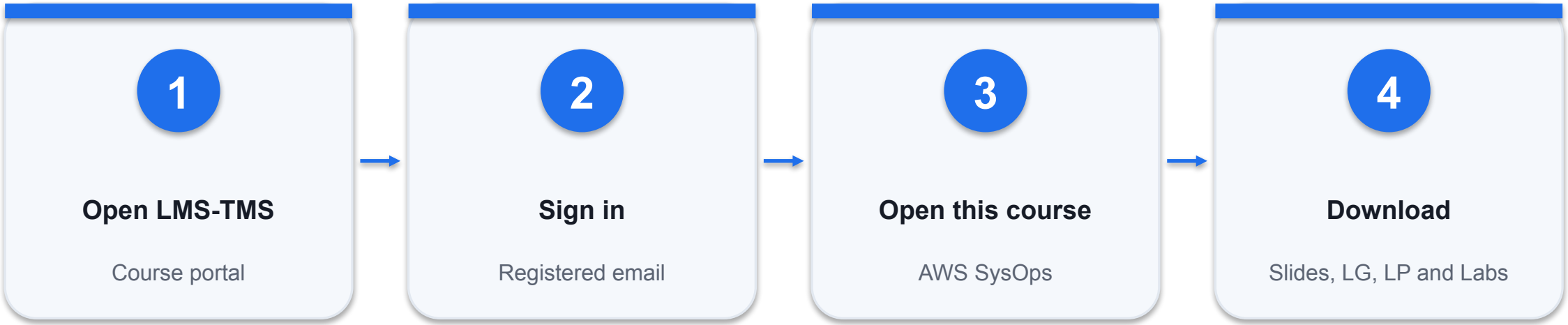
5. **Be punctual**
   Return on time and complete attendance.

6. **Clean up**
   Verify deletion of chargeable resources after every lab.

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VISUAL DOWNLOAD FLOW · LMS-TMS.TERTIARYINFOTECH.COM

Download Your Course Material



PROCEDURE BOUNDARY

Detailed lab procedures are in the Learner Guide and the individual lab folders.

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22 HOURS DELIVERY · 2 HOURS ASSESSMENT · 4 LABS/DAY

Lesson Plan — Three Days

1

Day 1

Foundations, observability, EC2 and fleet operations · Labs 01–04

2

Day 2

Reliability, continuity, deployment automation and security · Labs 05–08

3

Day 3

Networking, data operations, integrated troubleshooting and assessment · Labs 09–12

4

Assessment

WA-SAQ: 60 min · Case Study: 60 min

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WHAT YOU WILL BE ABLE TO DO

Learning Outcomes

1

LO1

Maintain business continuity of AWS cloud operations to ensure stable service delivery.

2

LO2

Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring.

3

LO3

Apply service quality standards to ensure security and compliance of AWS cloud operations.

4

LO4

Analyse and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.

5

LO5

Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.

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Skills Framework Alignment — Knowledge

1. **K1**
   Cloud computing infrastructure components and technologies
2. **K2**
   Installation, configuration and troubleshooting of cloud computing infrastructures
3. **K3**
   Installation, configuration and troubleshooting of software packages supporting cloud computing infrastructures
4. **K4**
   Design, configuration, maintenance and troubleshooting of cloud computing applications
5. **K5**
   Virtualisation technologies

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Skills Framework Alignment — Abilities

1

A1
Implement data loading, security, performance monitoring and tuning, capacity planning, backup and recovery

2

A2
Implement performance diagnostics and troubleshooting at scale in a live environment

3

A3
Maintain availability and performance of cloud computing operations and ensure secure services with no unplanned downtime

4

A4
Analyse processing issues and data anomalies and identify solutions

5

A5
Implement and release cloud computing operations changes

6

A6
Implement IT risk-management measures for business continuity and disaster recovery

7

A7
Implement service-quality standards

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CURRENT CLOUDOPS BLUEPRINT · 10 TOPICS · 12 LABS

Course Outline

1 **Topic 01 · Cloud Operations Foundations**
Shared responsibility · regions and zones · operational excellence · SOA-C03 transition

2 **Topic 02 · Monitoring, Logging and Incident Response**
CloudWatch · CloudTrail · EventBridge · alarms · anomaly detection · remediation

3 **Topic 03 · Compute, Virtualisation and Systems Management**
EC2 lifecycle · AMIs · placement · Systems Manager · patching · Image Builder

4 **Topic 04 · Reliability, Elasticity and High Availability**
Elastic Load Balancing · Auto Scaling · health checks · containers · graceful degradation

5 **Topic 05 · Backup, Restore and Business Continuity**
RTO/RPO · AWS Backup · snapshots · replication · pilot light · warm standby

6 **Topic 06 · Deployment, Provisioning and Automation**
CloudFormation · StackSets · change sets · drift · Systems Manager Automation · Git and IaC

7 **Topic 07 · Security, Identity and Compliance**
IAM · Organizations · KMS · Config · Security Hub · GuardDuty · Inspector · Macie

8 **Topic 08 · Networking, DNS and Content Delivery**
VPC · routes · security groups · NACLs · endpoints · Route 53 · CloudFront

9 **Topic 09 · Storage, Data and Database Operations**
S3 · EBS · EFS · RDS · DynamoDB · lifecycle · performance · data protection

10 **Topic 10 · Performance, Cost and Integrated Troubleshooting**
Bottleneck analysis · Cost Explorer · Trusted Advisor · runbooks · game days · capstone

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HANDS-ON CLOUD OPERATIONS

AWS Lab Environment

1

Learner account

Use only the account and Region authorised by the trainer.

2

AWS-native assets

Each folder includes editable YAML, JSON, CLI, shell or Python files.

3

Visual reference

Every lab includes an EXPECTED-ARCHITECTURE.jpg image.

4

Evidence

Retain outputs, decisions, verification and cleanup records.

5

Cost control

Use small resources, bounded tests and explicit cleanup.

6

Schedule

Exactly four labs are completed on each training day.

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WA-SAQ + CASE STUDY

Assessment Briefing

1

Individual work

No discussion, recording or sharing of assessment scripts.

2

Open book

Use slides, Learner Guide and approved materials only.

3

Written Assessment

Five open-ended questions · K1–K5 · 60 minutes.

4

Case Study

Five connected tasks · A1–A7 · 60 minutes.

5

Submission

Complete and upload the candidate papers through LMS-TMS.

6

Outcome

Competent or Not Yet Competent; reassessment and appeal rights apply.

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ON ASSESSMENT DAY

Assessment Flow

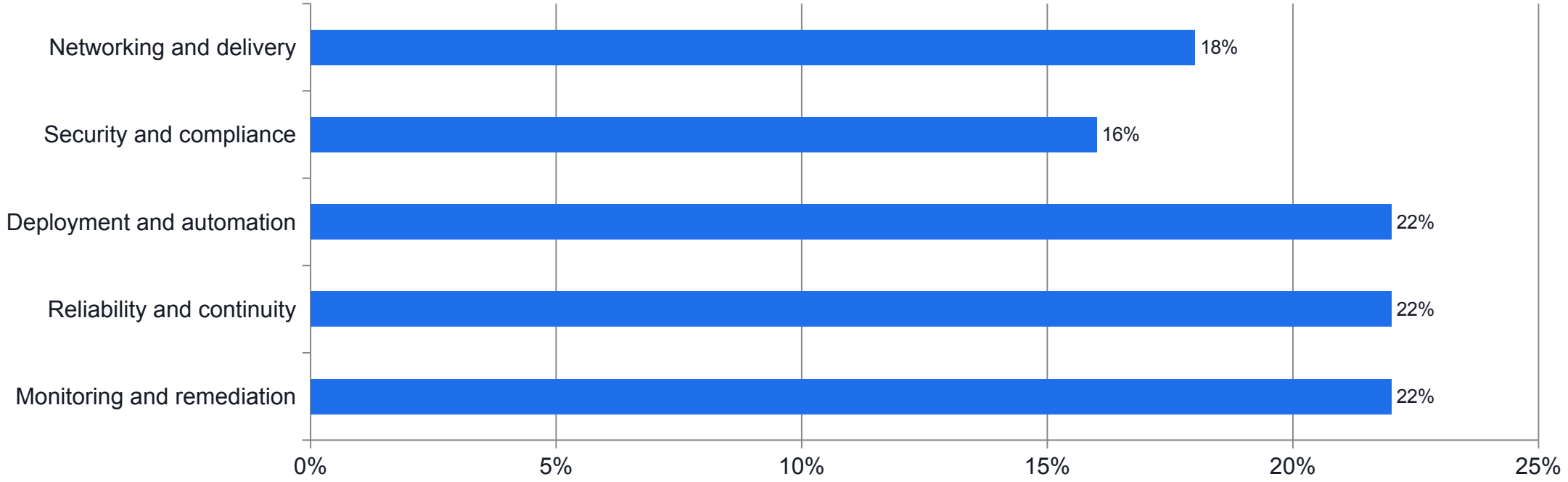
1. **TRAQOM**
   Course feedback
2. **Attendance**
   Assessment QR
3. **WA-SAQ**
   5 questions · 60 min
4. **Case Study**
   5 tasks · 60 min
5. **Submit**
   LMS-TMS and sign-off

CONFIDENTIALITY

Candidate papers are learner-facing; answer keys remain trainer-only.

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Current SOA-C03 Exam Domain Weighting



WHAT THE DATA SHOWS

Three domains carry equal 22% weight; the course integrates all five through observability, controlled change, recovery and troubleshooting.

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CURRENT AWS AUTHORITY · HTTPS://EXAMS.TERTIARYINFOTECH.COM

Certification Readiness

1

Current exam

AWS Certified CloudOps Engineer — Associate (SOA-C03).

2

Exam format

65 questions · 130 minutes · scaled passing score 720.

3

Role transition

The WSQ title remains SysOps; exam-readiness content follows the current CloudOps guide.

4

Practice Exam

Open exams.tertiaryinfotech.com and select the matching AWS CloudOps / SysOps practice exam; never use exam dumps.

5

Readiness loop

Explain the service choice, apply the labs, interpret evidence and justify recovery.

6

Scope

Current guide version 1.1, published 1 June 2026.

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VERSION v5.0 · 23 August 2026

Source and Version Authority

1. **WSQ authority**
   Live Tertiary course page and Assessment Plan v5.0 — Course code, 24-hour duration, five LO, K1–K5, A1–A7 and WA+CS design.

2. **Legacy content**
   Reference v3/v4 PowerPoints and Drive Activities dataset — Preserved useful EC2, SSM, CloudFormation, storage, monitoring, security and networking coverage; procedures moved out of the deck.

3. **Certification authority**
   AWS Certified CloudOps Engineer - Associate SOA-C03 Exam Guide v1.1 — Current five-domain blueprint and scope, including containers, multi-account operations, automation and IaC.

4. **Architecture guidance**
   AWS Well-Architected Framework and official AWS service documentation — Operational excellence, security, reliability, performance, cost and sustainability practices.

5. **Assessment authority**
   Drive-held WA/CS v2 papers and Assessment Plan v5.0 — Instrument type, five-item counts, 60-minute timings and controlled K/A coverage.

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01

TOPIC 01 · LO1 · LO2 · K1 · K5 · A1

Cloud Operations Foundations

Shared responsibility · regions and zones · operational excellence · SOA-C03 transition

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Key Ideas — Cloud Operations Foundations

1

Operate for outcomes

Availability, security, performance and cost are managed as measurable service outcomes.

2

Know the responsibility boundary

AWS secures the cloud; customers secure configurations, identities, data and workloads in the cloud.

3

Design for failure domains

Regions, Availability Zones and edge locations create distinct resilience and latency choices.

4

Use evidence before change

Metrics, logs, configuration history and events establish the operational baseline.

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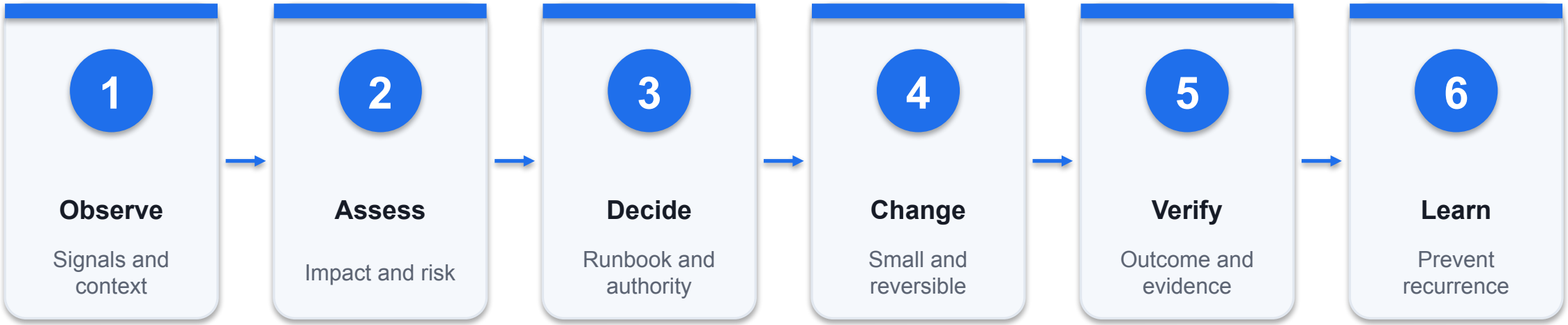
SysOps to CloudOps — What Changed

Area	Legacy SOA-C02 emphasis	Current SOA-C03 emphasis
Role name	SysOps Administrator	CloudOps Engineer
Workloads	EC2-heavy operations	Compute, containers, serverless and hybrid operations
Scope	Six broad domains	Five task-oriented domains
Automation	AWS-native operations	IaC, Git and third-party automation are explicit

WHEN IT MATTERS The WSQ course title and code remain unchanged; exam-readiness content follows the current SOA-C03 guide.

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The Cloud Operations Control Loop



PROFESSIONAL STANDARD

Every operational change must have a reason, owner, verification signal and rollback path.

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AWS Global Infrastructure

1

Region

Geographic boundary for sovereignty, latency and disaster-recovery decisions.

2

Availability Zone

Independent failure domain connected with low-latency networking.

3

Edge

CloudFront and Route 53 move delivery and routing closer to users.

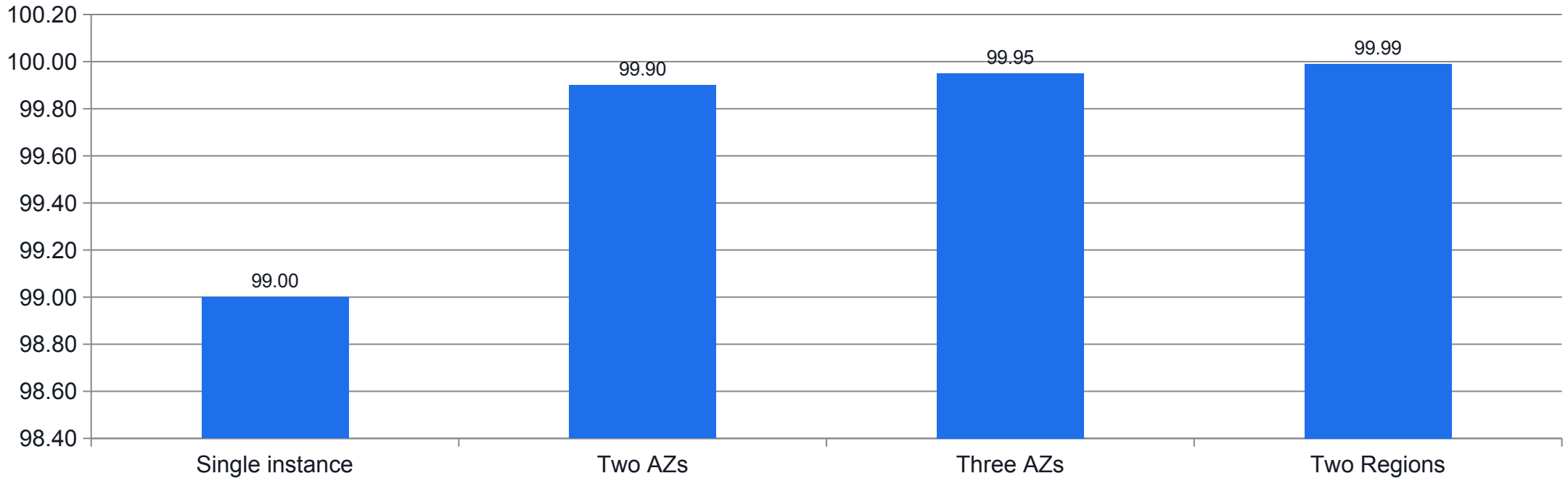
4

Local and hybrid

Outposts, Local Zones and connected environments extend the operating boundary.

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Availability Improves with Independent Failure Domains



WHAT THE DATA SHOWS

More failure domains increase potential resilience only when dependencies, health checks and failover are also designed correctly.

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Shared Responsibility by Layer

Layer	AWS responsibility	Customer responsibility
Facilities and hardware	Physical security, hosts and core network	Choose Regions and resilience strategy
Managed service	Service infrastructure and platform	Identity, configuration, data and use
EC2 workload	Underlying facilities and hypervisor	Guest OS, patches, application and traffic controls
Data	Durable service mechanisms	Classification, encryption, retention and access

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Well-Architected Operational Questions

Prepare

- What must be observable?
- What runbooks and permissions exist?

Operate

- What is normal?
- Who responds to deviation?

Evolve

- What did evidence teach?
- Which control prevents recurrence?

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Operations Mindset

The console is an interface—not the source of truth.

Treat configuration, telemetry, change history and verification evidence as the durable operational record.

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Recap — Cloud Operations Foundations

1

Explain

Availability, security, performance and cost are managed as measurable service outcomes.

2

Distinguish

AWS secures the cloud; customers secure configurations, identities, data and workloads in the cloud.

3

Apply

Use the control loop and responsibility boundary to frame an operational decision.

4

Evidence

Identify the configuration, telemetry, change and verification record required.

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02

TOPIC 02 · LO2 · LO4 · K1 · K3 · K4 · A1 · A4

Monitoring, Logging and Incident Response

CloudWatch · CloudTrail · EventBridge · alarms · anomaly detection · remediation

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Key Ideas — Monitoring, Logging and Incident Response

1

Metrics show behaviour

Dimensions, periods and statistics turn raw measurements into service signals.

2

Logs explain events

Structured records provide detail for search, correlation, retention and audit.

3

Traces connect dependencies

A request path exposes latency and failure across distributed components.

4

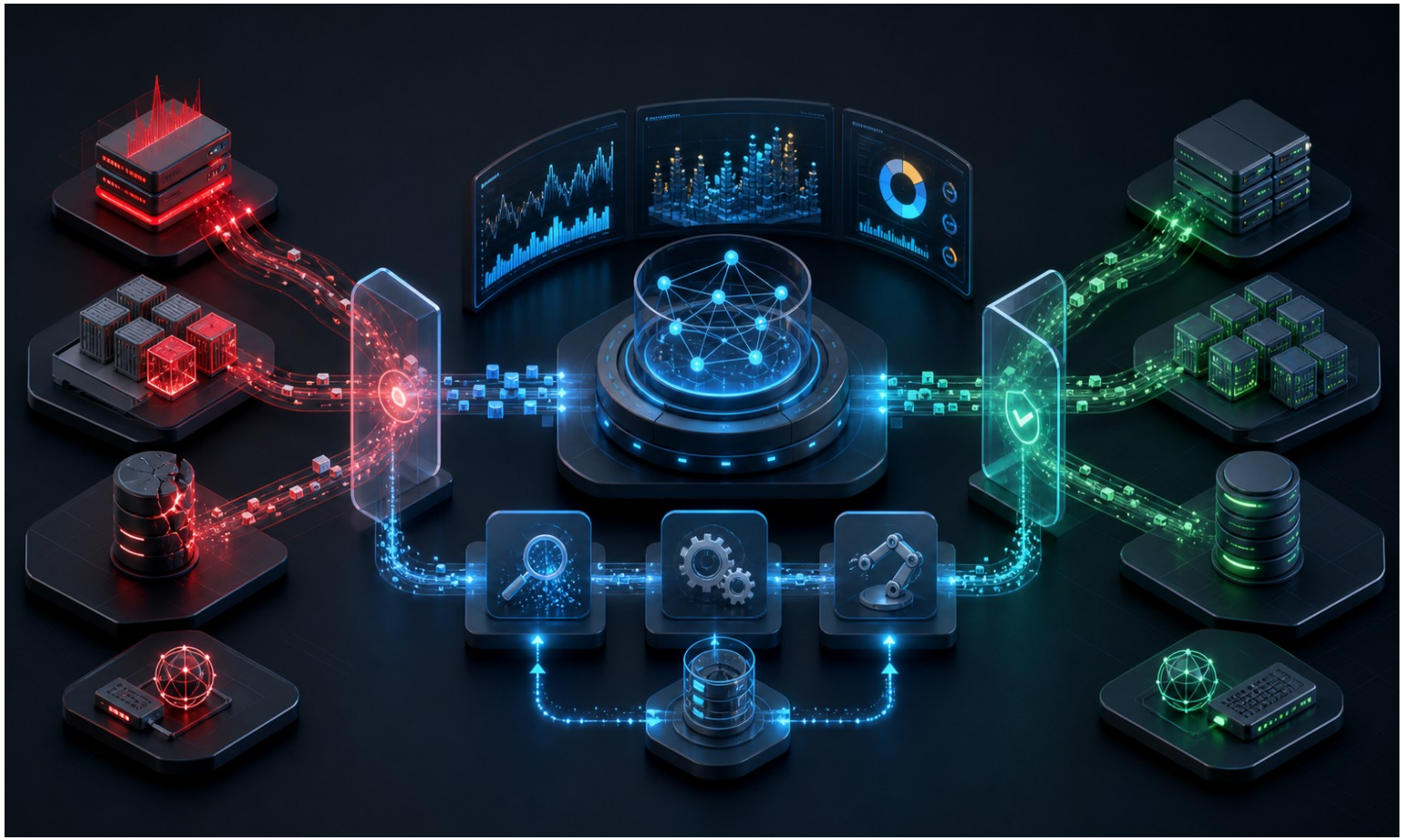
Incidents require controlled response

Detection, triage, containment, recovery and learning form one evidence chain.

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IMAGEGEN VISUAL · TOPIC 02

Observe, Decide, Remediate



READ THE SYSTEM

Read the scene as an operational control loop: signal, context, safe action, verification and escalation.

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Telemetry Pipeline



DESIGN TEST

A signal is useful only when it identifies impact, owner and next safe action.

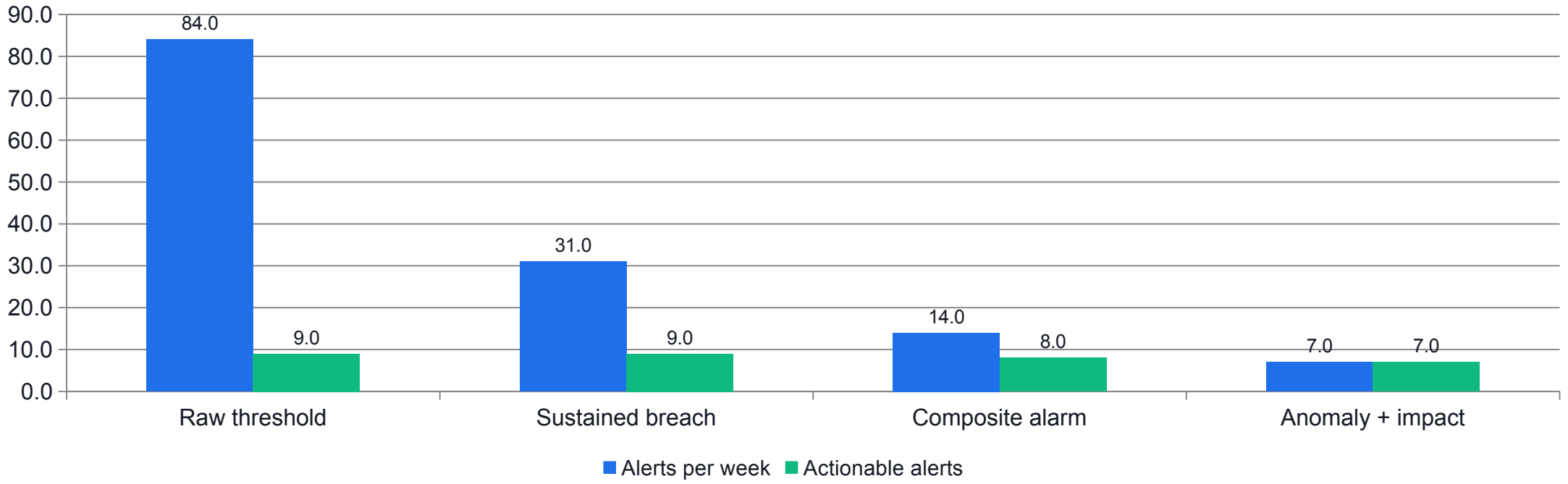
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CloudWatch, CloudTrail and Config

Service	Primary question	Typical evidence
CloudWatch	How is the workload behaving?	Metrics, logs, alarms and dashboards
CloudTrail	Who called which AWS API, when and from where?	Management and data events
AWS Config	What configuration existed and was it compliant?	Configuration history and rule evaluation

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Composite Signals Reduce Alert Noise



WHAT THE DATA SHOWS

Filtering transient and context-free signals can preserve actionable detection while reducing noise.

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Alarm Design

1

Statistic

Average can hide peaks; percentiles expose tail latency.

2

Period

Short periods detect quickly but may amplify noise.

3

Missing data

Choose breaching, not breaching, ignore or missing deliberately.

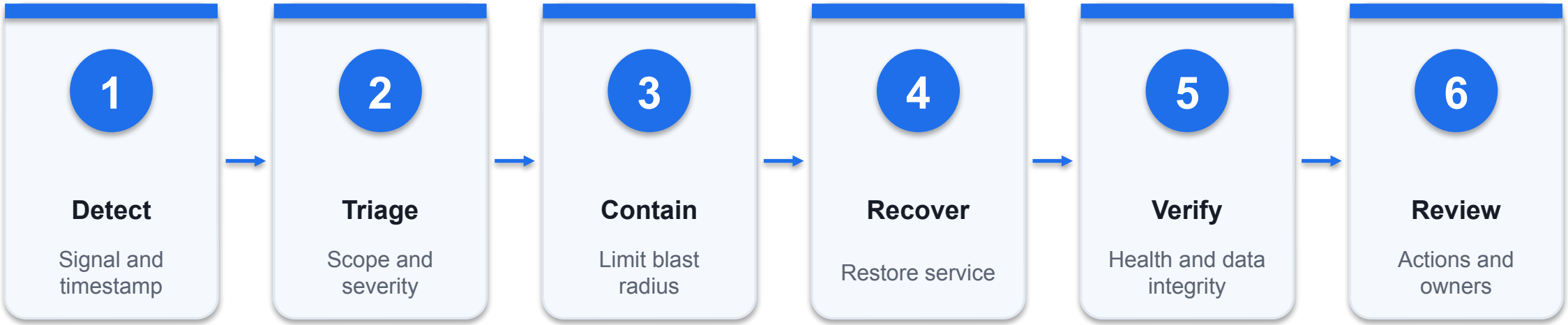
4

Action

Notify, scale or remediate only with tested permissions and rollback.

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Incident Lifecycle



A4 CONNECTION

Separate observed evidence, hypothesis, decision and result throughout the incident record.

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TOPIC 02 · K1 · K3 · K4 · A1 · A4

Threshold vs Anomaly Detection

Choice	Strength	Use when
Static threshold	Transparent and predictable	A meaningful limit is known
Anomaly detection	Adapts to time-varying baseline	Seasonality makes one limit noisy
Composite alarm	Combines symptom and impact	One signal alone is insufficient
Metric math	Derives rates and ratios	Raw counts need context

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Automation Guardrails

Trigger

- Use sustained evidence
- Prevent recursive events

Authority

- Least-privilege execution role
- Separate approval for high-risk action

Verification

- Check intended state
- Escalate when remediation fails

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TOPIC 02 · CONCEPT-LED LAB BRIEF

Build an Observability and Incident Detection Baseline

LAB 01

Apply K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.

Create measurable service signals, route one actionable event and record an evidence-led incident hypothesis.

YOU'LL BUILD

A CloudWatch dashboard, alarm definition, EventBridge pattern, evidence log and validation screenshots.

TOOLCHAIN

CloudWatch · CloudTrail · EventBridge · SNS · AWS CLI · CloudShell

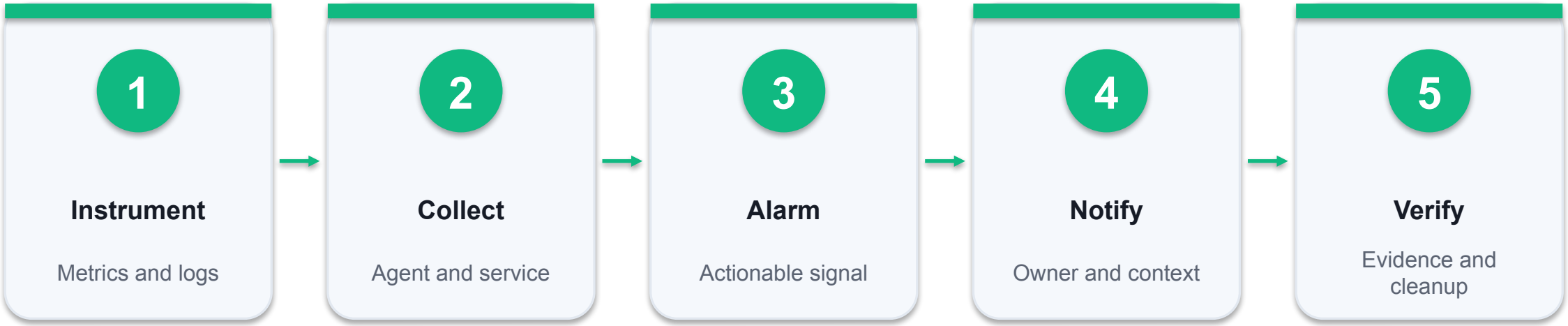
DONE WHEN

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

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LAB 01 · CONCEPTUAL WORKFLOW

Lab 01 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.

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LAB 01 · ACCEPTANCE EVIDENCE

Build an Observability and Incident Detection Baseline

✅ Expected result

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 02 · CONCEPT-LED LAB BRIEF

Automate Event-Driven Incident Triage

LAB 02

Apply K3, K4, A1 and A4 by routing a controlled AWS event through EventBridge, invoking a least-privilege remediation function and preserving an auditable triage record.

Build an event pattern and Lambda triage function that records evidence and recommends a bounded next action.

YOU'LL BUILD

An EventBridge rule, Lambda triage function, test event, CloudWatch Logs evidence and incident decision record.

TOOLCHAIN

EventBridge · Lambda · CloudTrail · CloudWatch Logs · IAM

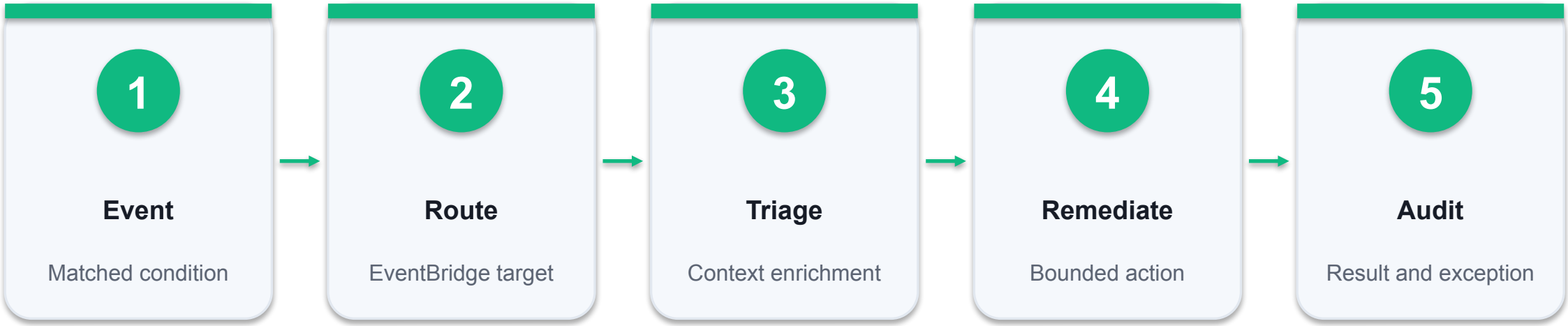
DONE WHEN

The test and real events reach the Lambda target, logs contain the event ID and actor context, no automatic network modification occurs, the decision record names authority and verification, and validation emits LAB 02 PASS.

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LAB 02 · CONCEPTUAL WORKFLOW

Lab 02 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.

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LAB 02 · ACCEPTANCE EVIDENCE

Automate Event-Driven Incident Triage

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Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Monitoring, Logging and Incident Response

1

Explain

Dimensions, periods and statistics turn raw measurements into service signals.

2

Distinguish

Structured records provide detail for search, correlation, retention and audit.

3

Apply

Apply K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.

4

Evidence

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

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03

TOPIC 03 · LO1 · LO2 · K3 · K5 · A3 · A5

Compute, Virtualisation and Systems Management

EC2 lifecycle · AMIs · placement · Systems Manager · patching · Image Builder

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Key Ideas — Compute, Virtualisation and Systems Management

1

Treat instances as managed nodes

Inventory, patch, access and automate consistently across a fleet.

2

Images are controlled supply-chain inputs

Build, scan, test and version AMIs instead of hand-configuring servers.

3

Placement affects failure and performance

Spread, partition and cluster strategies solve different operational needs.

4

Lifecycle actions need coordination

Launch, stop, terminate and scale transitions require hooks and evidence.

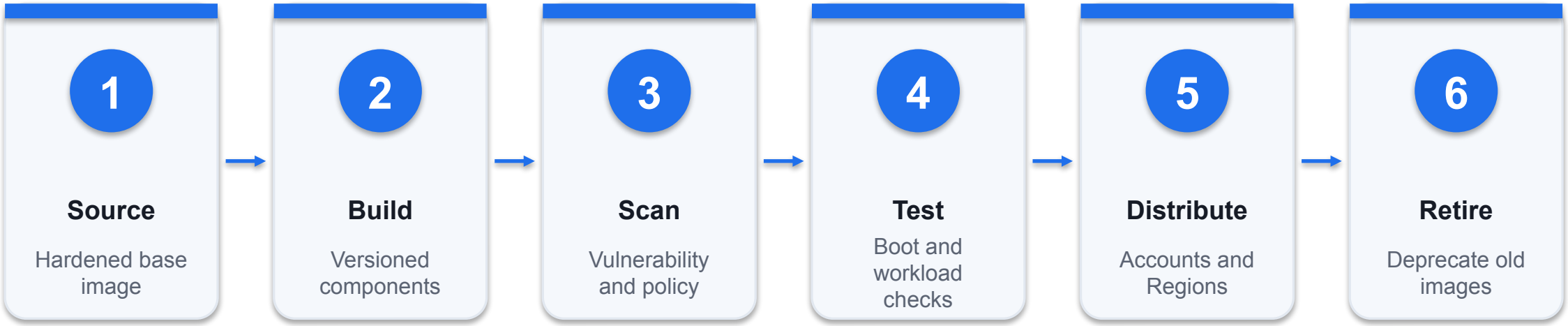
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EC2 Purchase and Capacity Models

Model	Operational value	Risk / constraint
On-Demand	Flexible baseline and unpredictable workloads	Higher unit cost
Savings Plans	Commitment discount across eligible usage	Forecast and commitment risk
Reserved capacity	Capacity assurance in an AZ	Scope and utilisation management
Spot	Large discount for interruptible work	Two-minute interruption and capacity variability

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Golden Image Pipeline

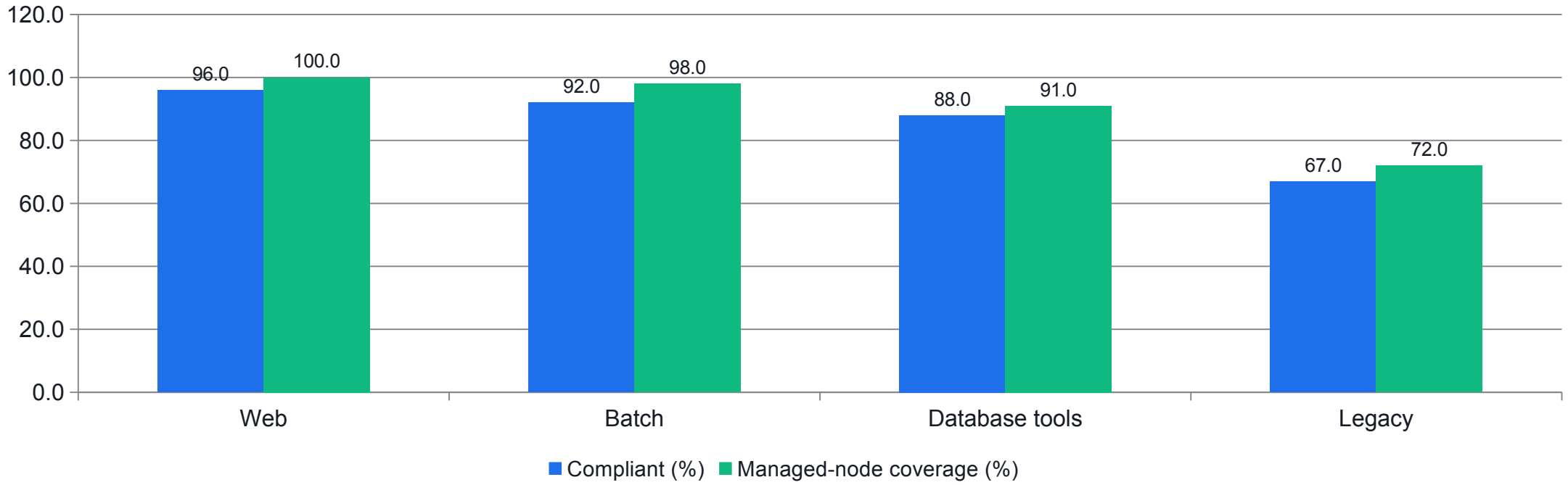


A5 CONTROL

An image release is an operational change with evidence, version and rollback.

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Patch Compliance by Fleet Segment



WHAT THE DATA SHOWS

Patch compliance cannot exceed managed-node visibility; onboard and inventory the fleet before enforcing targets.

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Systems Manager Capabilities

1

Session Manager

Auditable shell access without inbound SSH or bastion exposure.

2

Run Command

Execute controlled commands across tagged targets.

3

Patch Manager

Baseline, maintenance window, compliance and remediation.

4

Automation

Runbooks coordinate multi-step operational changes.

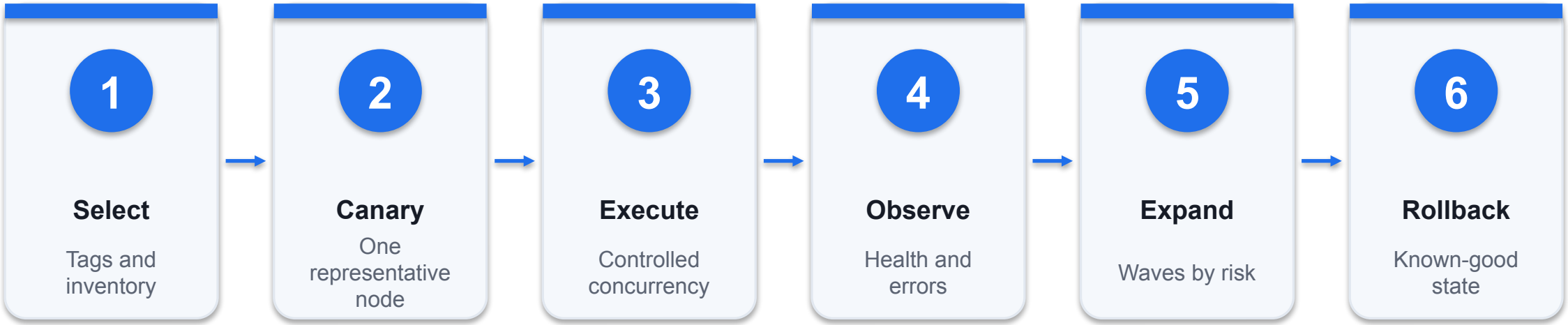
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Placement Strategies

Strategy	Optimises	Failure characteristic
Cluster	Low-latency high-throughput east-west traffic	Single-AZ concentration
Spread	Small number of critical instances	Distinct underlying hardware
Partition	Large distributed workloads	Partition-aware failure isolation

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Safe Fleet Change



OPERATIONS RULE

Scale the change only after the canary evidence meets the acceptance threshold.

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Fleet Principle

If a server cannot be identified, accessed, patched and replaced consistently, it is not operationally managed.

Use tags, roles, inventory, image pipelines and Systems Manager to turn instances into a governed fleet.

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TOPIC 03 · CONCEPT-LED LAB BRIEF

Operate an EC2 Fleet with Systems Manager

LAB 03

Apply K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled Automation runbook.

Build a least-privilege managed-node baseline and produce patch, command and change evidence without exposing inbound administration ports.

YOU'LL BUILD

An EC2 managed node, SSM document, patch-compliance report, automation execution record and rollback evidence.

TOOLCHAIN

EC2 · IAM · Systems Manager · Session Manager · Run Command · Patch Manager · Automation

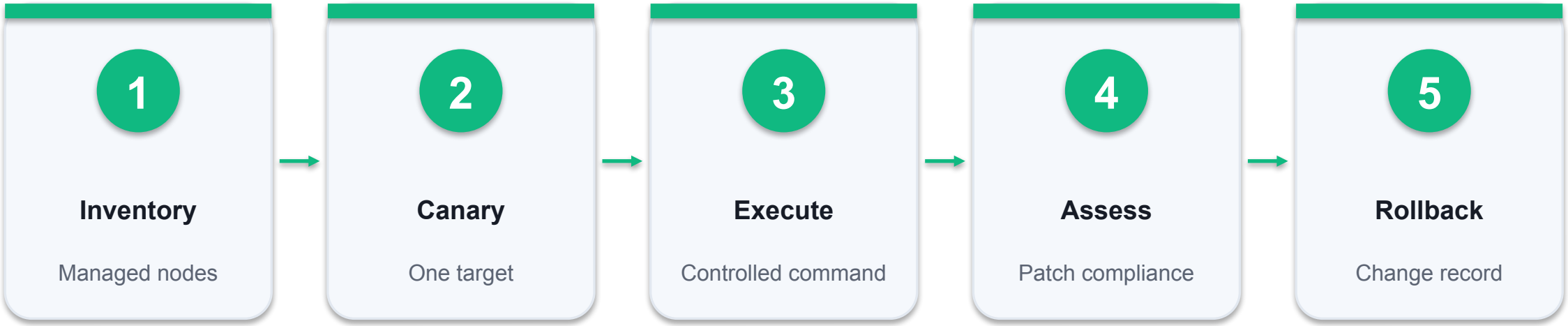
DONE WHEN

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

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LAB 03 · CONCEPTUAL WORKFLOW

Lab 03 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.

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LAB 03 · ACCEPTANCE EVIDENCE

Operate an EC2 Fleet with Systems Manager

✅ Expected result

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 03 · RETRIEVAL PRACTICE

Recap — Compute, Virtualisation and Systems Management

1

Explain

Inventory, patch, access and automate consistently across a fleet.

2

Distinguish

Build, scan, test and version AMIs instead of hand-configuring servers.

3

Apply

Apply K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled Automation runbook.

4

Evidence

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

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04

TOPIC 04 · LO1 · K4 · K5 · A3 · A5

Reliability, Elasticity and High Availability

Elastic Load Balancing · Auto Scaling · health checks · containers · graceful degradation

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Key Ideas — Reliability, Elasticity and High Availability

1

High availability is an end-to-end property

Compute redundancy fails if identity, data or network dependencies remain single points.

2

Elasticity follows demand

Auto Scaling changes capacity using a measurable signal and bounded policy.

3

Health checks drive routing

A check must represent the user's ability to receive a correct response.

4

Resilience requires recovery

Design degradation, retry, failover and rollback before an incident.

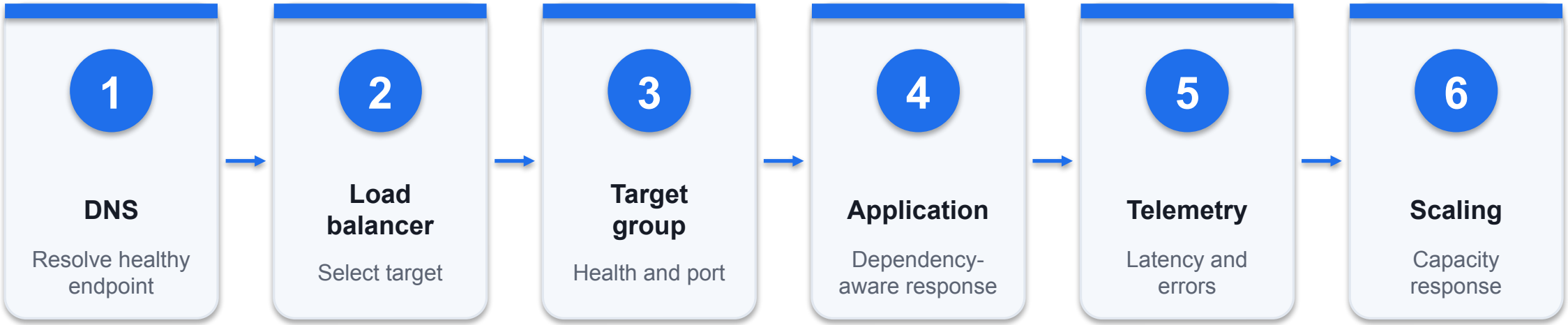
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Load Balancer Selection

Type	Layer and strength	Typical workload
ALB	Layer 7 HTTP routing, host/path rules	Web, APIs and containers
NLB	Layer 4 performance and static IP	TCP/UDP/TLS and extreme scale
GWLB	Transparent appliance insertion	Firewalls and inspection fleets

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Request Path and Health

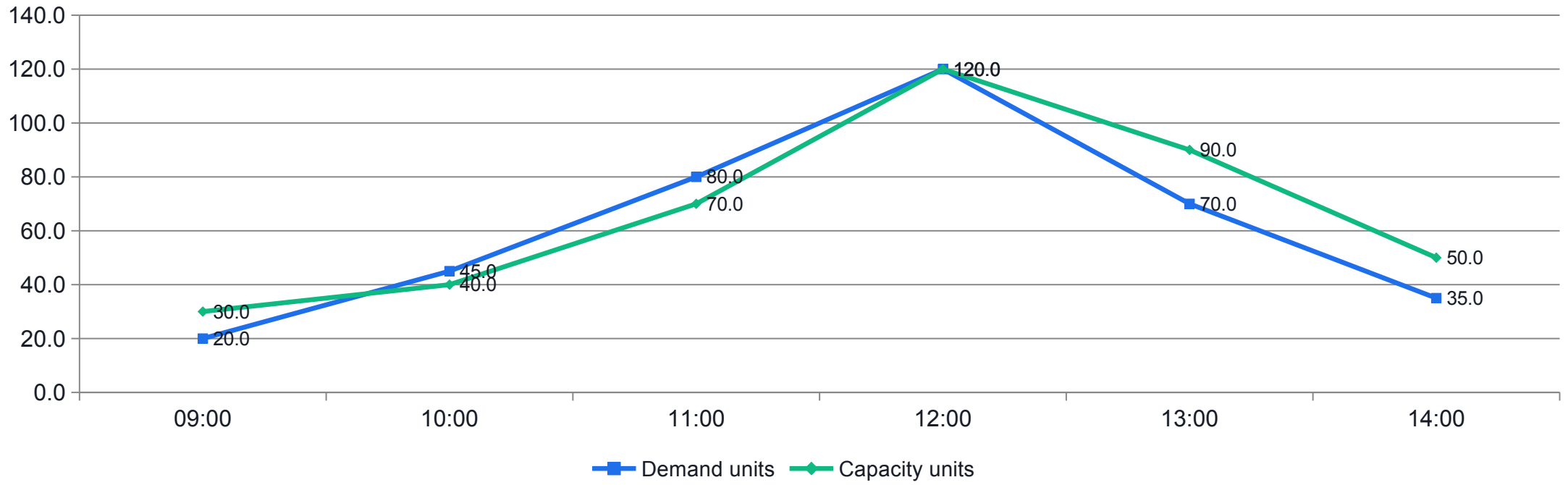


TROUBLESHOOTING

Follow the request path in order; do not jump directly to the instance.

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Scaling Policy Behaviour



WHAT THE DATA SHOWS

Target tracking follows demand with delay; warm-up and cooldown prevent unstable oscillation.

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Auto Scaling Controls

1

Desired capacity

The current fleet target between minimum and maximum bounds.

2

Target tracking

Adjusts capacity to hold a metric near a chosen target.

3

Step scaling

Uses breach magnitude to change capacity by defined steps.

4

Lifecycle hook

Pauses launch or termination for controlled work.

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TOPIC 04 · K4 · K5 · A3 · A5

Failure Handling

Pattern	Response	Caution
Retry with backoff	Recover from transient dependency failure	Bound attempts and add jitter
Circuit breaker	Stop calling an unhealthy dependency	Define recovery probe
Bulkhead	Isolate resource pools	Capacity must reflect priority
Graceful degradation	Preserve critical function	Communicate reduced service

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TOPIC 04 · K4 · K5 · A3 · A5

Container Operations in SOA-C03

ECS / EKS

- Service desired count
- Task or pod health and placement

Images

- Versioned and scanned
- Immutable deployment input

Observability

- Container Insights
- Logs, metrics and traces by workload

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Availability Test

Redundancy is not resilience until failover is exercised.

Measure detection, decision, traffic shift, recovery, data integrity and return to steady state.

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TOPIC 04 · CONCEPT-LED LAB BRIEF

Deploy and Test a Multi-AZ Highly Available Service

LAB 04

Apply K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a controlled change.

Deploy a small Multi-AZ web tier, verify health and scaling signals, then run a bounded failure exercise.

YOU'LL BUILD

A two-AZ VPC web tier, ALB, target group, launch template, Auto Scaling group, failover evidence and cleanup record.

TOOLCHAIN

VPC · EC2 · ALB · Auto Scaling · CloudWatch · CloudFormation

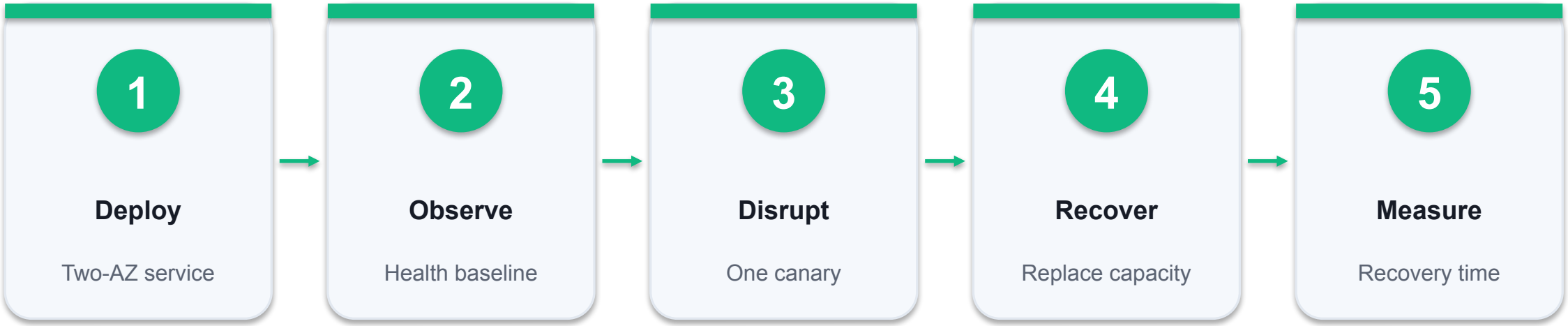
DONE WHEN

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

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LAB 04 · CONCEPTUAL WORKFLOW

Lab 04 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.

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LAB 04 · ACCEPTANCE EVIDENCE

Deploy and Test a Multi-AZ Highly Available Service

✅ Expected result

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Reliability, Elasticity and High Availability

1

Explain

Compute redundancy fails if identity, data or network dependencies remain single points.

2

Distinguish

Auto Scaling changes capacity using a measurable signal and bounded policy.

3

Apply

Apply K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a controlled change.

4

Evidence

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

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05

TOPIC 05 · LO1 · LO4 · K4 · A1 · A4 · A6

Backup, Restore and Business Continuity

RTO/RPO · AWS Backup · snapshots · replication · pilot light · warm standby

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Key Ideas — Backup, Restore and Business Continuity

1

Backup is not recovery

A protected copy has value only when restore time and data integrity are proven.

2

RTO and RPO drive design

Recovery time and acceptable data loss determine technology and cost.

3

Cross-account protection limits blast radius

Separate vault ownership and controls protect against compromised workloads.

4

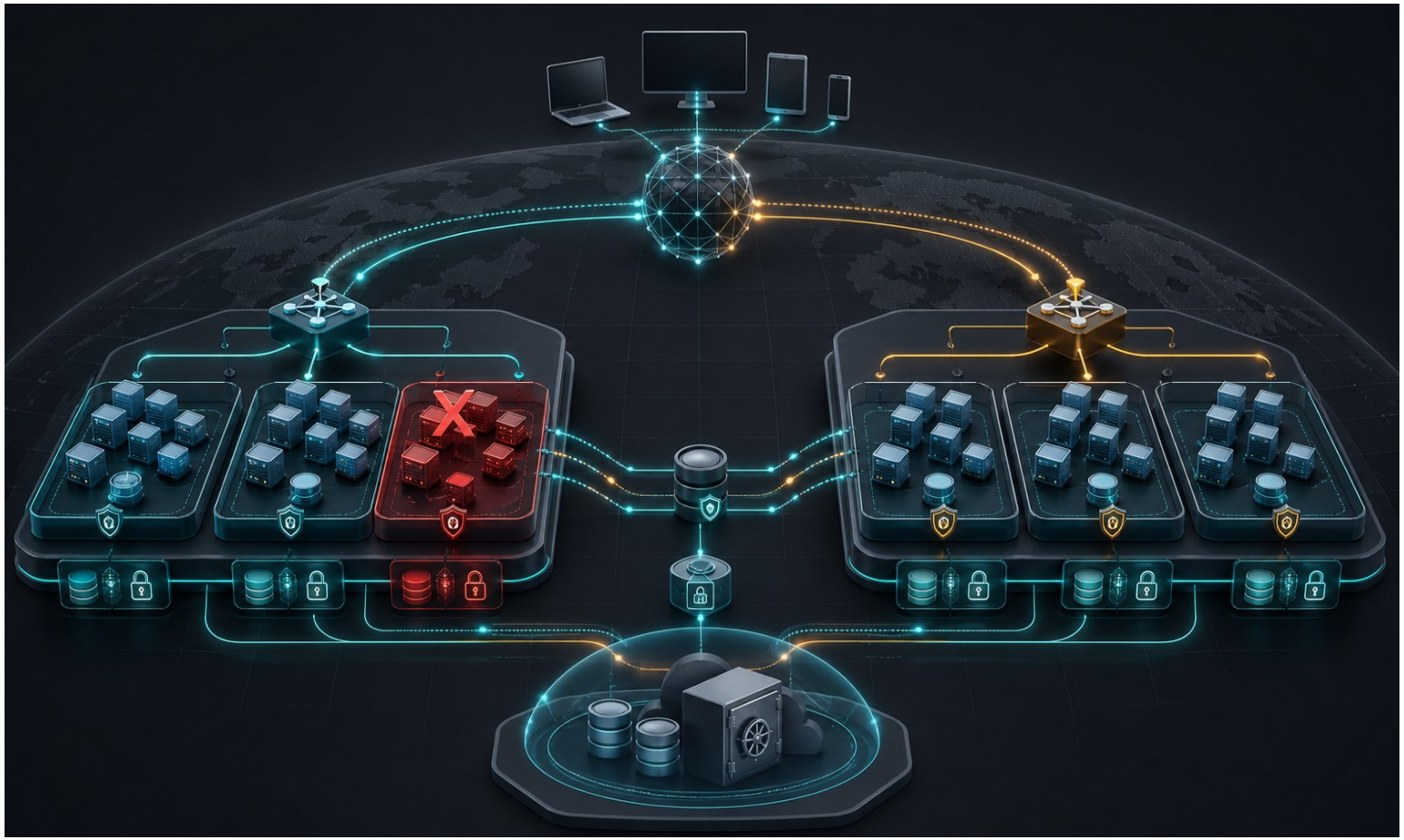
Continuity is a rehearsed capability

People, runbooks, dependencies and evidence matter as much as replication.

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IMAGEGEN VISUAL · TOPIC 05

Resilience Is a Tested Capability



READ THE SYSTEM

Identify independent failure domains, recovery paths and the evidence needed to prove continuity.

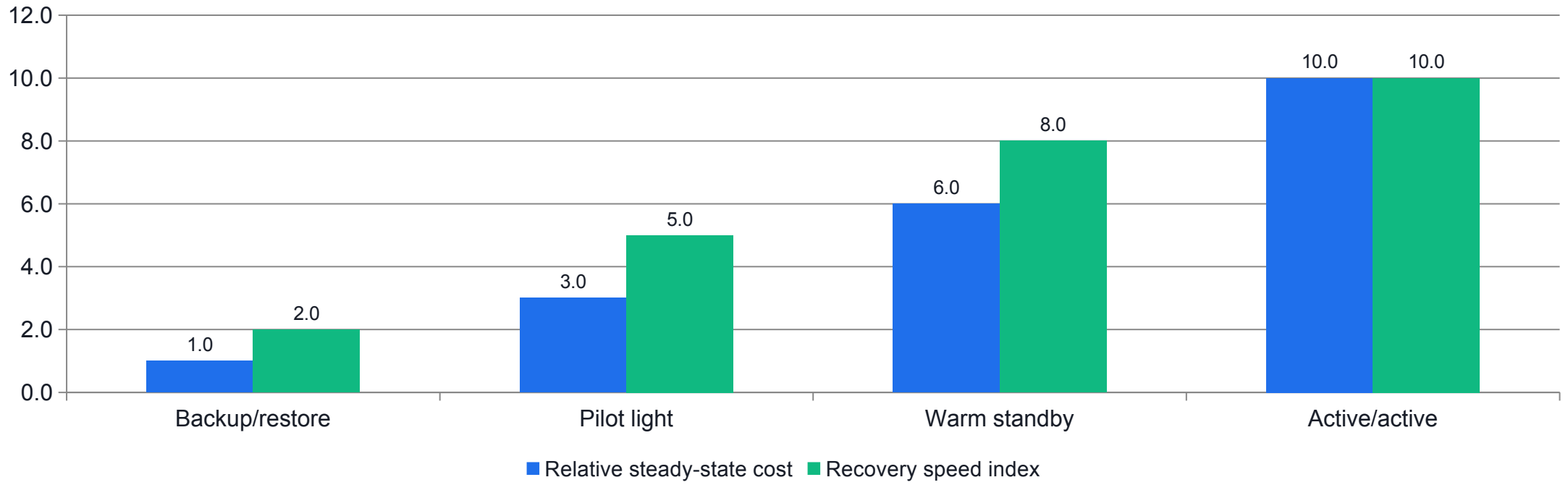
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RTO and RPO

Objective	Question	Design influence
RTO	How quickly must service return?	Automation, standby capacity and failover
RPO	How much recent data may be lost?	Backup frequency and replication
Retention	How far back must recovery reach?	Lifecycle, vault and compliance
Integrity	How do we know restored data is correct?	Checksums and application validation

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Continuity Patterns Trade Cost for Recovery

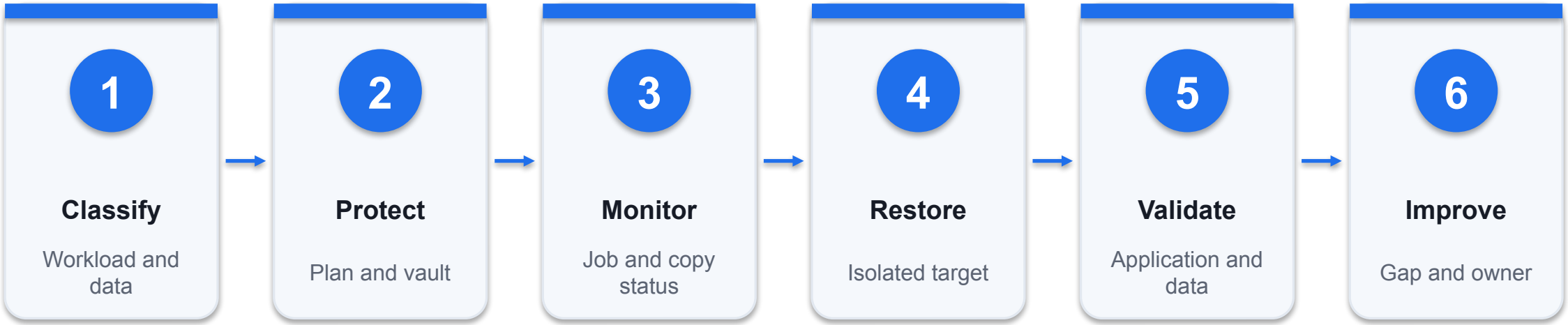


WHAT THE DATA SHOWS

Choose the least costly pattern that demonstrably meets business RTO, RPO and compliance requirements.

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Recovery Assurance Loop



A6 EVIDENCE

A completed backup job is an input; a tested restore with measured recovery is evidence.

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AWS Backup Controls

1

Plan

Schedules, windows, lifecycle and copy actions.

2

Vault

Access policy, encryption and logical separation.

3

Vault Lock

Retention controls designed to resist early deletion.

4

Audit Manager

Evidence against backup policy and compliance frameworks.

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Replication Choices

Mechanism	Strength	Operational concern
EBS snapshot copy	Point-in-time regional protection	Restore and launch workflow
S3 replication	Object-level asynchronous copy	Existing objects and delete markers
RDS cross-Region replica	Lower RPO read copy	Promotion and endpoint change
DynamoDB global table	Multi-Region active data	Conflict and application behaviour

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Continuity Runbook

Decision

- Declared trigger and authority
- Communication path

Execution

- Dependency order
- Automated and manual steps

Evidence

- Timestamps and validation
- Actual RTO/RPO and gaps

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Recovery Principle

A restore you have not tested is only a hypothesis.

Schedule recovery exercises, isolate tests from production, measure actual results and close every gap with an owner.

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TOPIC 05 · CONCEPT-LED LAB BRIEF

Prove Backup and Restore Readiness

LAB 06

Apply K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.

Create a backup plan and vault configuration, run a protected copy, restore into isolation and validate integrity and recovery time.

YOU'LL BUILD

A backup plan, selection, restore checklist, recovery evidence and measured RTO/RPO report.

TOOLCHAIN

AWS Backup · S3 · EBS · KMS · CloudWatch Events

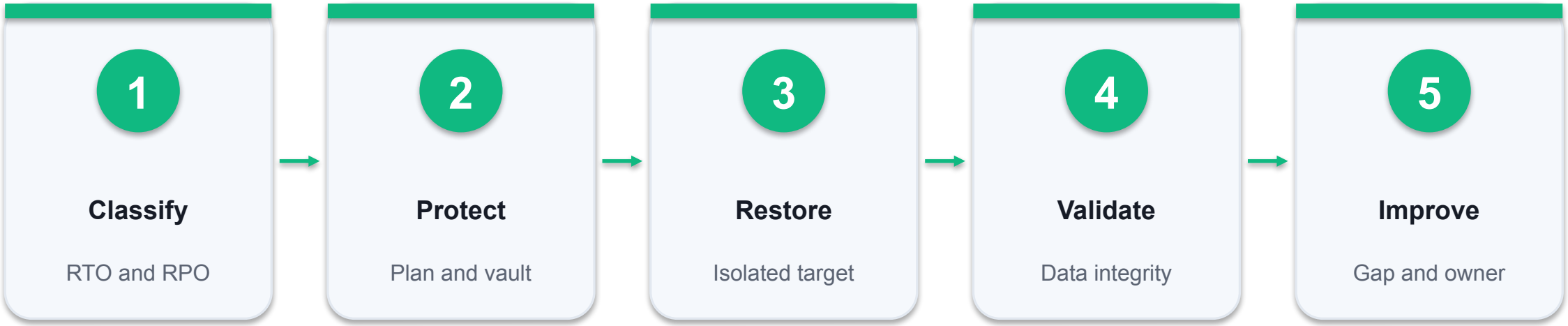
DONE WHEN

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

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LAB 06 · CONCEPTUAL WORKFLOW

Lab 06 Evidence Model



PROCEDURE BOUNDARY

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LAB 06 · ACCEPTANCE EVIDENCE

Prove Backup and Restore Readiness

✅ Expected result

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Backup, Restore and Business Continuity

1

Explain

A protected copy has value only when restore time and data integrity are proven.

2

Distinguish

Recovery time and acceptable data loss determine technology and cost.

3

Apply

Apply K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.

4

Evidence

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

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06

TOPIC 06 · LO2 · LO4 · K3 · K4 · A4 · A5

Deployment, Provisioning and Automation

CloudFormation · StackSets · change sets · drift · Systems Manager Automation · Git and IaC

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Key Ideas — Deployment, Provisioning and Automation

1

Infrastructure is versioned

Templates and policies create a reviewable desired state.

2

Preview before execution

Change sets reveal creates, replacements and deletions before the stack changes.

3

Drift is a control signal

Difference between desired and actual state requires investigation, not blind overwrite.

4

Automation must be idempotent

Repeated execution should converge safely to the intended state.

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Infrastructure Change Path



A5 CONNECTION

The template, plan, approval, execution and verification form one auditable release record.

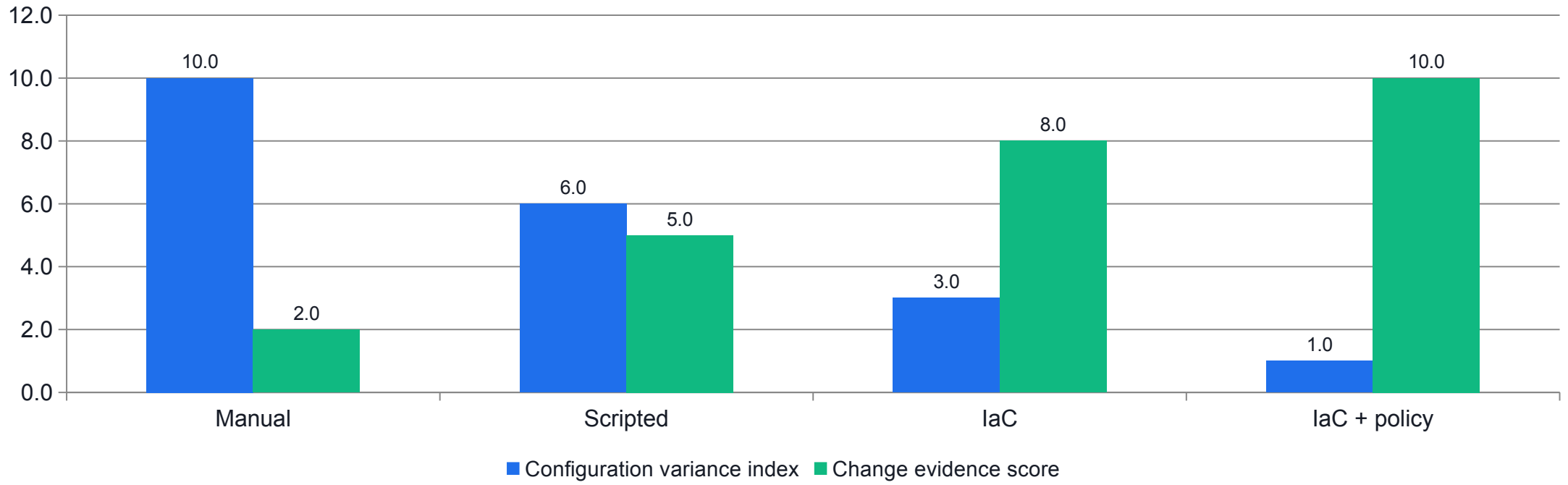
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CloudFormation Update Controls

Control	Purpose	Operational use
Change set	Preview resource actions	Review replacements and deletions
Stack policy	Protect critical resources	Restrict updates during normal change
Rollback	Return after failed update	Preserve events for diagnosis
Termination protection	Prevent accidental stack deletion	Use on persistent environments

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Automation Reduces Configuration Variance



WHAT THE DATA SHOWS

Automation improves consistency; policy and version control make the change explainable and governable.

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Template Engineering

1

Parameters

Expose safe environment-specific values without duplicating templates.

2

Mappings and conditions

Represent controlled variation, not arbitrary complexity.

3

Outputs and exports

Publish stable integration values with dependency awareness.

4

Deletion and update policies

Protect or snapshot stateful resources during change.

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StackSets Deployment Models

Model	Best fit	Trade-off
Self-managed	Explicit administration and target roles	More role setup and control
Service-managed	AWS Organizations integration	Requires trusted access and OU planning
Automatic deployment	New accounts inherit baseline	Test policy and Region scope carefully

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Automation Runbook Design

Inputs

- Validated parameters
- No embedded secrets

Control

- Retries and timeouts
- Approval for destructive steps

Output

- Structured evidence
- Explicit success and rollback

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IaC Discipline

Do not fix drift by hiding it.

Identify who changed what and why, decide whether actual or desired state is correct, then reconcile through the controlled source.

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TOPIC 06 · CONCEPT-LED LAB BRIEF

Control Infrastructure Change and Configuration Drift

LAB 05

Apply K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation runbook with verification.

Use desired-state evidence to identify drift, choose the correct state, reconcile through code and verify the application outcome.

YOU'LL BUILD

A validated CloudFormation stack, reviewed change set, drift report, reconciliation decision, Automation verification and cleanup record.

TOOLCHAIN

CloudFormation · Change Sets · Drift Detection · Systems Manager Automation · AWS CLI

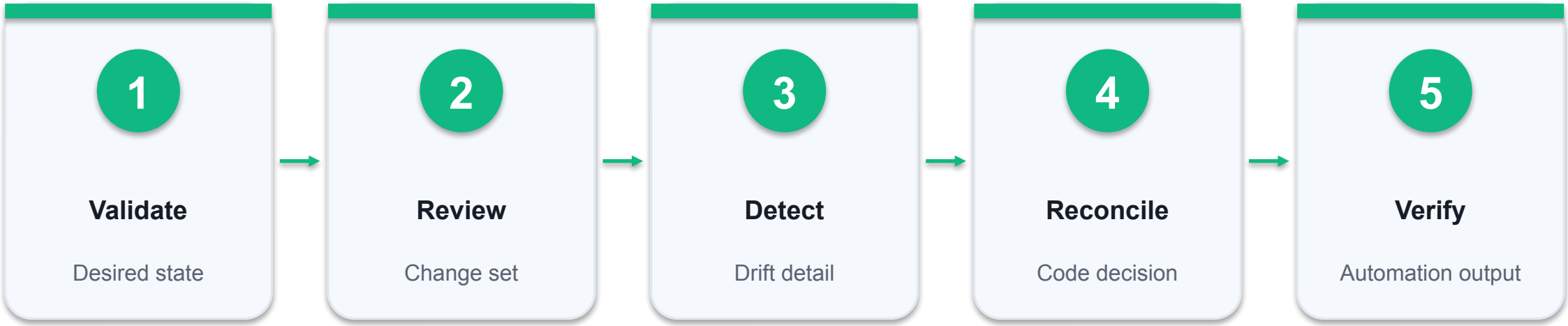
DONE WHEN

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

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LAB 05 · CONCEPTUAL WORKFLOW

Lab 05 Evidence Model



PROCEDURE BOUNDARY

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LAB 05 · ACCEPTANCE EVIDENCE

Control Infrastructure Change and Configuration Drift

✅ Expected result

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 06 · RETRIEVAL PRACTICE

Recap — Deployment, Provisioning and Automation

1

Explain

Templates and policies create a reviewable desired state.

2

Distinguish

Change sets reveal creates, replacements and deletions before the stack changes.

3

Apply

Apply K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation runbook with verification.

4

Evidence

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

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07

TOPIC 07 · LO2 · LO3 · K1 · K3 · A1 · A7

Security, Identity and Compliance

IAM · Organizations · KMS · Config · Security Hub · GuardDuty · Inspector · Macie

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Key Ideas — Security, Identity and Compliance

1

Identity is the control plane

Short-lived credentials, least privilege and explicit trust bound administrative power.

2

Prevent, detect and respond

Policies prevent known bad states; telemetry detects deviation; runbooks contain and recover.

3

Encryption includes key operations

Access, rotation, grants, aliases and recovery must be governed—not only enabled.

4

Compliance requires evidence

A rule result is useful when linked to scope, owner, remediation and exception process.

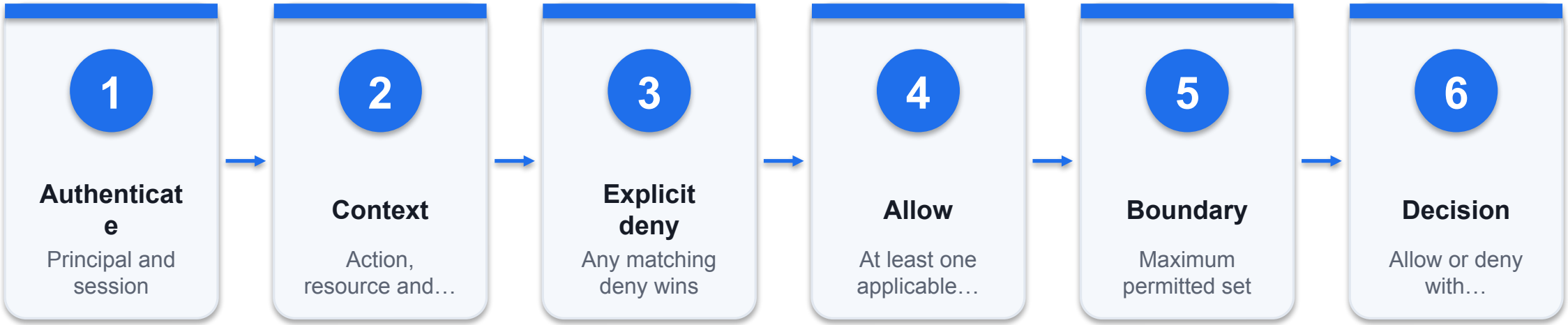
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Identity Policy Layers

Layer	What it controls	Common failure
Identity policy	Permissions granted to a principal	Overbroad action or resource
Resource policy	Who may access a resource	Unintended external principal
Permission boundary	Maximum identity permissions	Assuming it grants access
SCP	Maximum permissions in an organisation	Blocking required service operation

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Access Evaluation

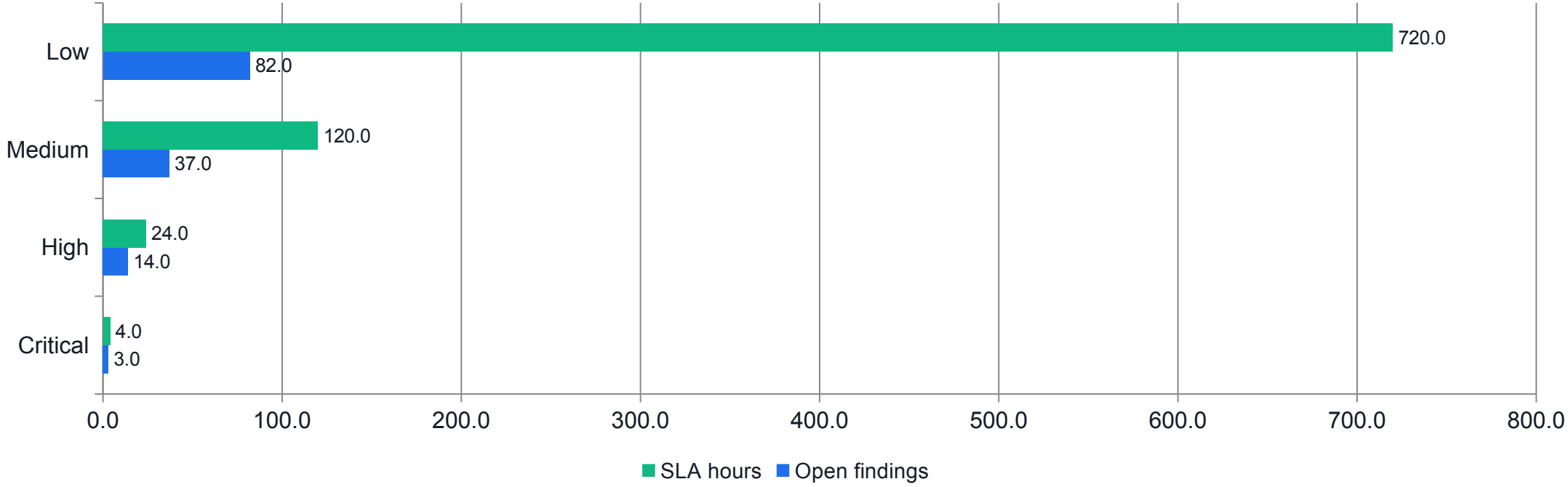


TROUBLESHOOTING

Use CloudTrail, policy simulator and access analysis to explain the decision path.

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Security Findings Need Prioritisation



WHAT THE DATA SHOWS

Severity, exposure, asset value and exploitability drive response; high volume alone does not determine priority.

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Security Services by Question

1

GuardDuty

What suspicious account, workload or network activity is occurring?

2

Inspector

Which compute workloads have vulnerabilities or exposure?

3

Macie

Where is sensitive data in S3 and how is it exposed?

4

Security Hub

How do findings and standards combine across accounts?

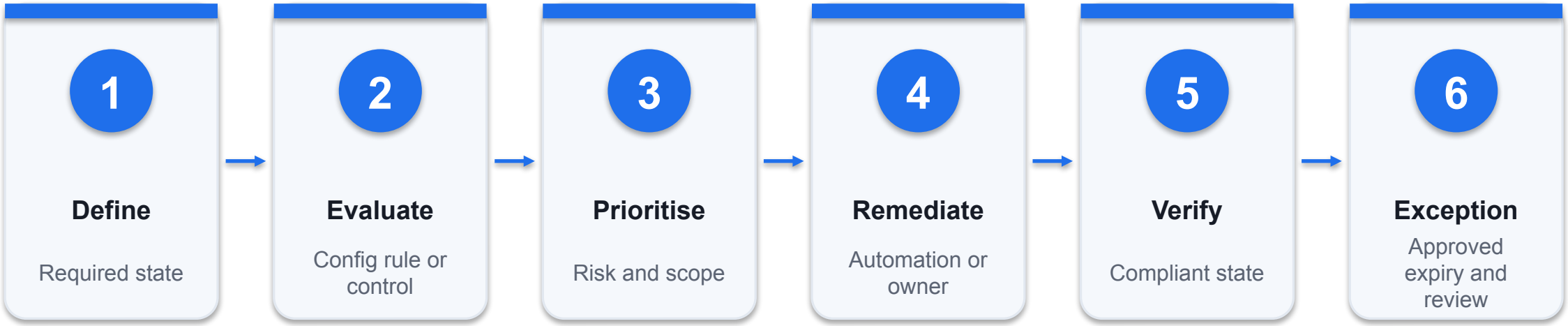
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KMS Key Choices

Choice	Operational control	Use carefully
AWS owned	Service manages lifecycle	Limited customer control
AWS managed	Visible service key	Policy is service-oriented
Customer managed	Policy, grants, rotation and aliases	Availability and deletion governance
Multi-Region key	Related key material across Regions	Application and recovery design

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Compliance Remediation



A7 OUTPUT

Service quality is a measurable standard with evidence, not a one-time checkbox.

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Security Operations

Least privilege is a lifecycle, not a launch-day setting.

Observe actual use, remove unused permissions, constrain trust, investigate denied actions and revalidate after organisational change.

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TOPIC 07 · CONCEPT-LED LAB BRIEF

Build a Multi-Account Compliance Evidence Flow

LAB 07

Apply K1, K3, A1 and A7 by evaluating required tags and encryption, aggregating findings and documenting a controlled remediation exception.

Deploy Config-style controls, review findings and produce auditable remediation and exception evidence.

YOU'LL BUILD

A Config rule template, tag policy, findings register, remediation decision and service-quality scorecard.

TOOLCHAIN

AWS Config · Organizations · Security Hub · EventBridge · Systems Manager Automation

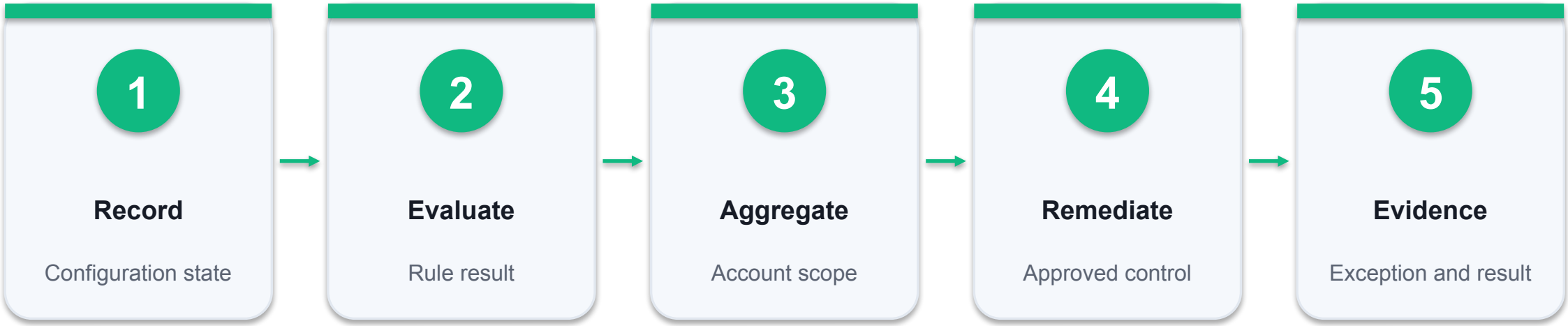
DONE WHEN

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.

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LAB 07 · CONCEPTUAL WORKFLOW

Lab 07 Evidence Model



PROCEDURE BOUNDARY

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LAB 07 · ACCEPTANCE EVIDENCE

Build a Multi-Account Compliance Evidence Flow

✅ Expected result

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 07 · CONCEPT-LED LAB BRIEF

Diagnose IAM Access and Protect Data with KMS

LAB 08

Apply K1, K3, A1 and A7 by explaining an access decision, narrowing permissions and verifying encryption and key-policy behaviour.

Use policy evidence to distinguish identity, resource and KMS decisions, then implement and verify least privilege.

YOU'LL BUILD

A policy-evaluation worksheet, corrected least-privilege policy, KMS key-policy review and encrypted-object evidence.

TOOLCHAIN

IAM · Access Analyzer · Policy Simulator · KMS · S3 · CloudTrail

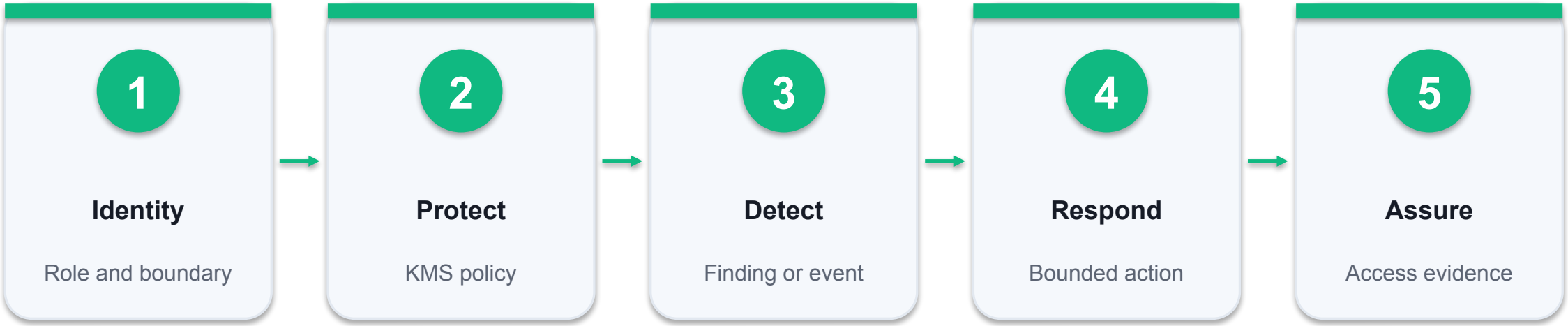
DONE WHEN

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.

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LAB 08 · CONCEPTUAL WORKFLOW

Lab 08 Evidence Model



PROCEDURE BOUNDARY

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LAB 08 · ACCEPTANCE EVIDENCE

Diagnose IAM Access and Protect Data with KMS

✅ Expected result

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Security, Identity and Compliance

1

Explain

Short-lived credentials, least privilege and explicit trust bound administrative power.

2

Distinguish

Policies prevent known bad states; telemetry detects deviation; runbooks contain and recover.

3

Apply

Apply K1, K3, A1 and A7 by evaluating required tags and encryption, aggregating findings and documenting a controlled remediation exception.

4

Evidence

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.

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08

TOPIC 08 · LO5 · K2 · A2

Networking, DNS and Content Delivery

VPC · routes · security groups · NACLs · endpoints · Route 53 · CloudFront

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Key Ideas — Networking, DNS and Content Delivery

1

Connectivity is a path

DNS, route, policy, state, endpoint and application health must all succeed.

2

Security groups are stateful

Return traffic is automatically allowed; network ACLs evaluate each direction.

3

Private connectivity reduces exposure

VPC endpoints, Transit Gateway and hybrid links change routing and policy boundaries.

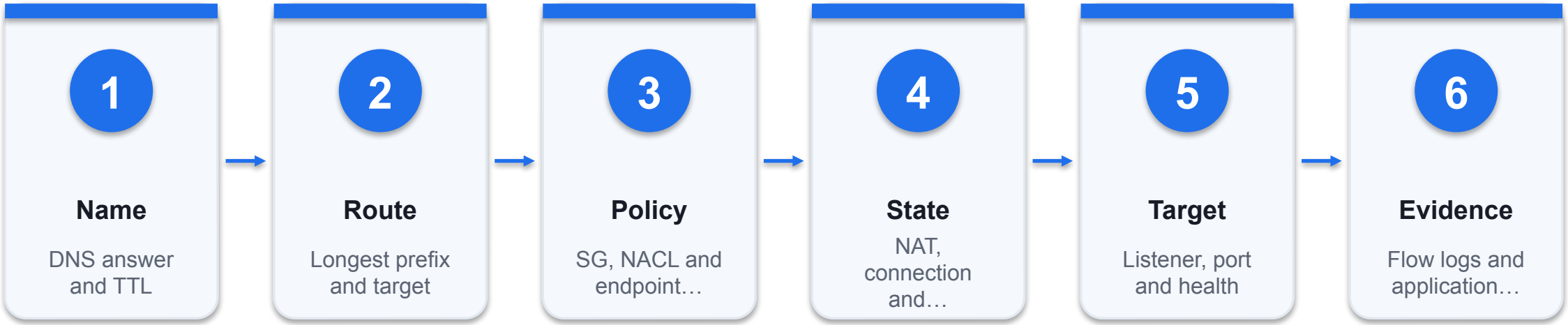
4

DNS is a control surface

Routing policy, health evaluation, TTL and resolver behaviour affect recovery.

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Connectivity Troubleshooting Order



A2 STANDARD

Test from source to destination and record the first failed control boundary.

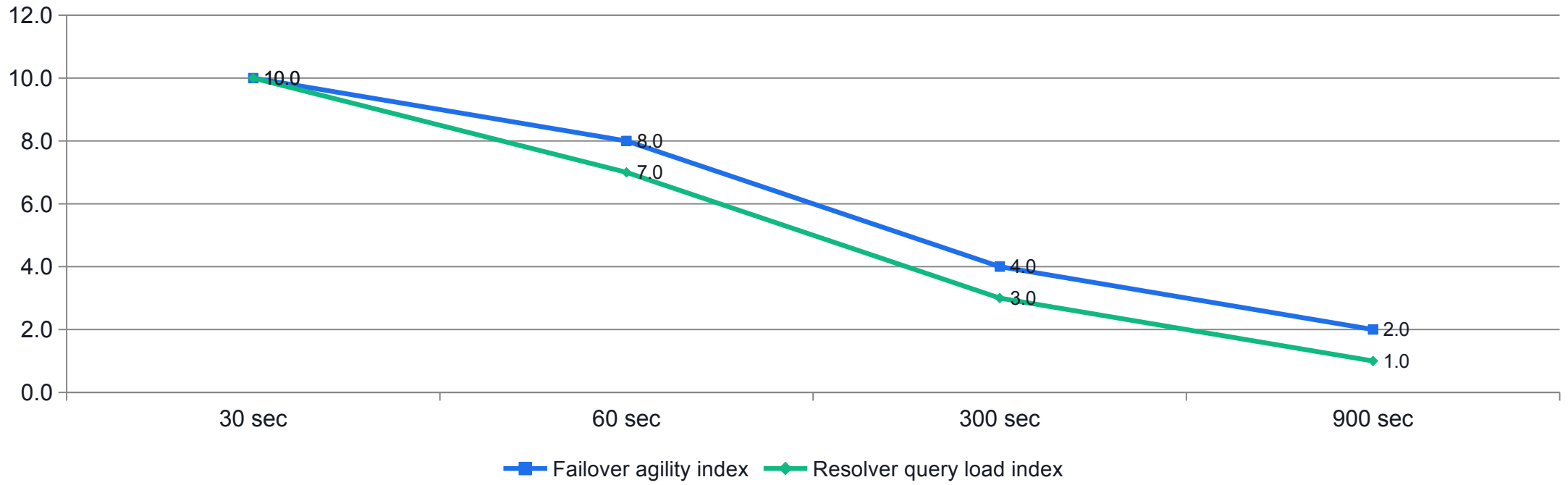
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Security Group vs Network ACL

Property	Security group	Network ACL
Scope	Elastic network interface	Subnet
State	Stateful	Stateless
Rules	Allow only	Numbered allow and deny
Evaluation	All rules together	Lowest rule number first

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DNS TTL Trades Agility for Query Load



WHAT THE DATA SHOWS

Lower TTL can speed client refresh but increases query volume and cannot force every resolver to behave identically.

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Route 53 Routing Policies

1

Simple

One or more values without health-based policy logic.

2

Weighted

Controlled traffic split for migration or experiments.

3

Latency

Direct users to the Region with lowest measured latency.

4

Failover

Primary and secondary routing driven by health.

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VPC Endpoint Types

Endpoint	Connectivity	Common use
Gateway	Route-table target	S3 and DynamoDB
Interface	Private IP through PrivateLink	Many AWS and partner services
Gateway Load Balancer	Inspection appliance insertion	Central security services
Resource / service network	VPC Lattice private application access	Cross-VPC service connectivity

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CloudFront Operations

Cache

- Cache key and TTL
- Invalidation and versioned objects

Origin

- Origin access control
- Origin health and failover

Security

- TLS certificate and WAF
- Logs and real-time metrics

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Network Diagnosis

A timeout is a symptom, not a location.

Use DNS results, route tables, reachability analysis, flow logs, load-balancer health and application evidence to isolate the boundary.

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TOPIC 08 · CONCEPT-LED LAB BRIEF

Troubleshoot a VPC Connectivity Path

LAB 09

Apply K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.

Diagnose the first failed control boundary using a repeatable evidence ladder and implement one verified correction.

YOU'LL BUILD

A VPC diagnostic worksheet, reachability/flow-log evidence, corrected network control and before/after result.

TOOLCHAIN

VPC · Reachability Analyzer · Flow Logs · Security Groups · NACLs · Endpoints

DONE WHEN

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

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LAB 09 · CONCEPTUAL WORKFLOW

Lab 09 Evidence Model



PROCEDURE BOUNDARY

Detailed commands and numbered steps are in the Learner Guide and lab folder—not on these slides.

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LAB 09 · ACCEPTANCE EVIDENCE

Troubleshoot a VPC Connectivity Path

✅ Expected result

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

IF IT DOESN'T WORK

Input mismatch

Reconcile identifiers, required fields and units with the supplied data dictionary.

Validation stop

Inspect the named rule and source row; preserve the check rather than bypassing it.

Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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TOPIC 08 · CONCEPT-LED LAB BRIEF

Operate DNS and Content-Delivery Failover

LAB 10

Apply K2 and A2 by configuring health-aware DNS and CloudFront behaviour, then proving client-visible routing and cache outcomes.

Model primary/secondary routing, inspect health status and TTL, then verify CloudFront origin and cache evidence.

YOU'LL BUILD

A Route 53 health-check definition, failover record plan, CloudFront operations checklist and measured failover report.

TOOLCHAIN

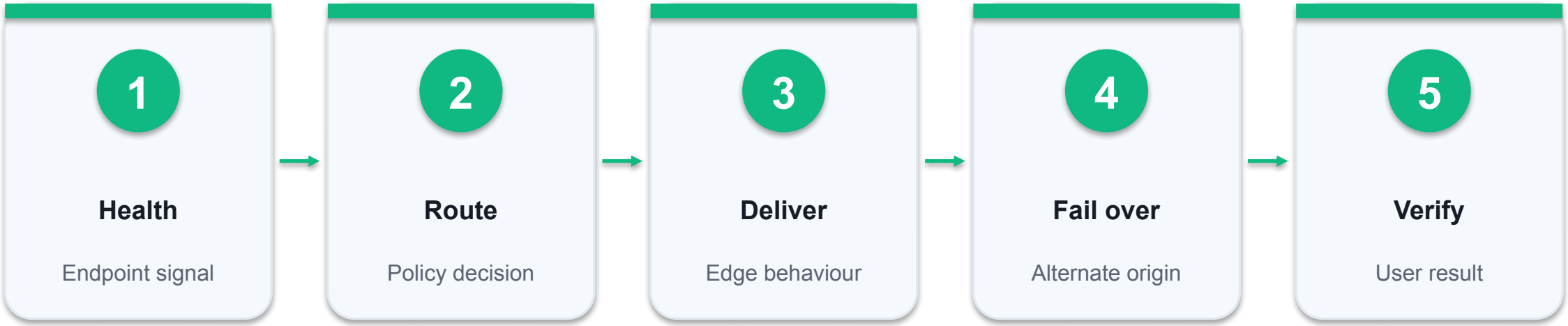
Route 53 · Health Checks · CloudFront · S3 · CloudWatch

DONE WHEN

Health and record evidence are linked, failover timing is measured rather than assumed, cache and DNS effects are distinguished, baseline is restored, and validation emits LAB 10 PASS.

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Lab 10 Evidence Model



PROCEDURE BOUNDARY

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LAB 10 · ACCEPTANCE EVIDENCE

Operate DNS and Content-Delivery Failover

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Recap — Networking, DNS and Content Delivery

1

Explain

DNS, route, policy, state, endpoint and application health must all succeed.

2

Distinguish

Return traffic is automatically allowed; network ACLs evaluate each direction.

3

Apply

Apply K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.

4

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The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

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09

TOPIC 09 · LO1 · LO2 · LO4 · K1 · K4 · A1 · A4 · A6

Storage, Data and Database Operations

S3 · EBS · EFS · RDS · DynamoDB · lifecycle · performance · data protection

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Key Ideas — Storage, Data and Database Operations

1

Choose by access pattern

Object, block, file and database services solve different consistency, latency and sharing needs.

2

Performance has service-specific limits

IOPS, throughput, connections, partitions and caching require different evidence.

3

Data protection is layered

Encryption, versioning, snapshots, backup and replication address different risks.

4

Managed does not mean unmanaged

Customers still own schema, configuration, access, capacity and recovery choices.

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Storage Selection

Service	Access model	Operational fit
S3	Object and API	Durable data lake, backup and static content
EBS	Block device to compute	Low-latency instance and database volumes
EFS	Shared NFS file system	Elastic multi-instance Linux file access
FSx	Managed specialised file system	Windows, Lustre, NetApp or OpenZFS workloads

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S3 Data Protection

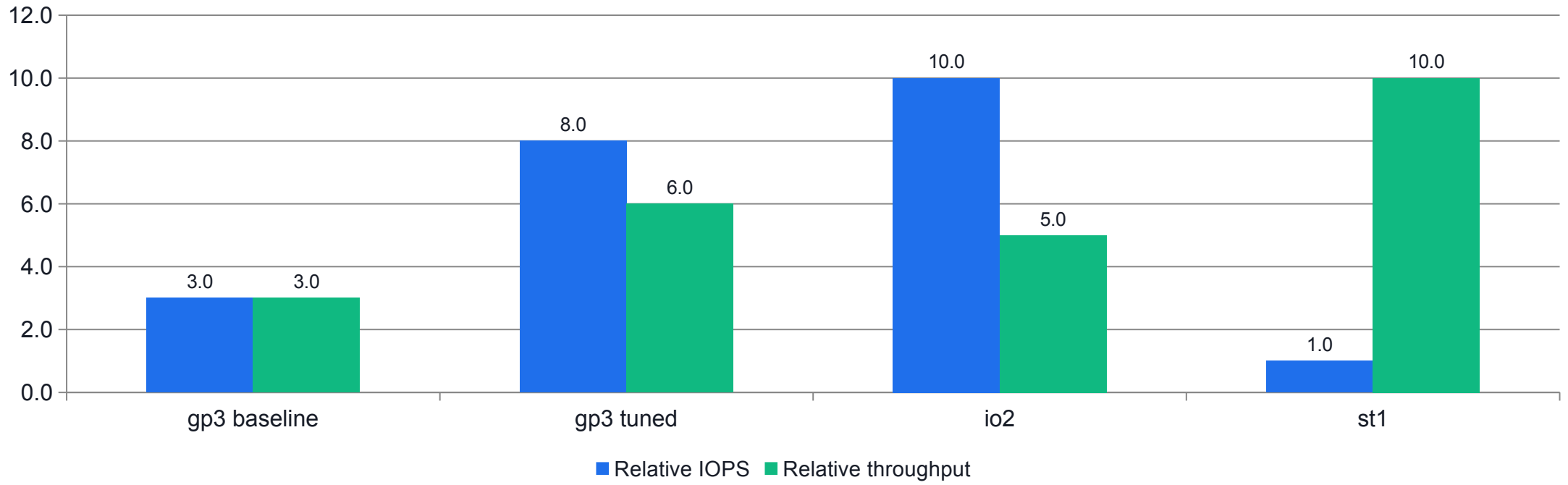


A1 CONNECTION

Protection combines preventive access, recoverability, retention and evidence.

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EBS Performance Must Match Workload



WHAT THE DATA SHOWS

IOPS and throughput are different dimensions; select volume type and configuration from measured workload behaviour.

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RDS Availability and Scale

Feature	Primary goal	Operational note
Multi-AZ	Availability and failover	Standby is not a read-scaling target
Read replica	Read scale and reporting	Replication lag and promotion plan
Automated backup	Point-in-time recovery	Retention and restore testing
RDS Proxy	Connection pooling and resilience	Application compatibility and limits

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DynamoDB Operations

1

Capacity

On-demand or provisioned with auto scaling and adaptive capacity.

2

Partition key

Distribute traffic; avoid hot keys and unbounded item growth.

3

Recovery

Point-in-time recovery, on-demand backup and restore.

4

Global table

Multi-Region active access with application-aware conflict behaviour.

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Database Incident Evidence

Resource

- CPU, memory and storage
- Connections and queue depth

Engine

- Slow queries and locks
- Wait events and plans

Change

- Configuration and schema history
- Deployment and failover events

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Data Operations

Durability claims do not replace recovery design.

Protect the correct scope, control access, monitor jobs, test restore, validate application behaviour and measure actual RTO/RPO.

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TOPIC 09 · CONCEPT-LED LAB BRIEF

Diagnose Storage and Database Performance, Then Recover

LAB 11

Apply K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.

Analyse a synthetic metric set, choose one evidence-based action and verify a point-in-time recovery plan.

YOU'LL BUILD

A metric analysis, bottleneck decision, change plan, restore checklist and database recovery evidence.

TOOLCHAIN

CloudWatch · EBS · RDS · Performance Insights · AWS Backup

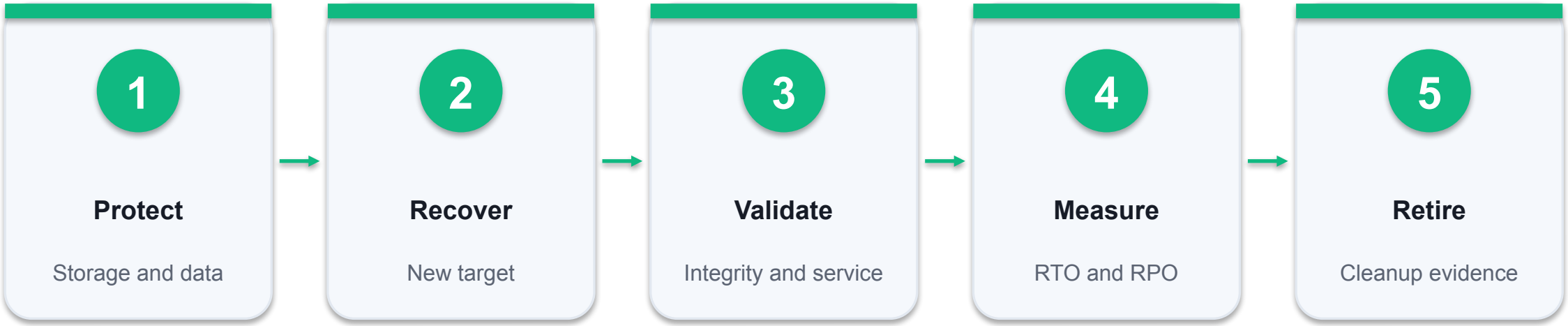
DONE WHEN

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.

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LAB 11 · CONCEPTUAL WORKFLOW

Lab 11 Evidence Model



PROCEDURE BOUNDARY

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LAB 11 · ACCEPTANCE EVIDENCE

Diagnose Storage and Database Performance, Then Recover

✅ Expected result

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.

IF IT DOESN'T WORK

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Reconcile identifiers, required fields and units with the supplied data dictionary.

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Evidence differs

Check the denominator, time window and assumptions before rerunning from clean inputs.

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Recap — Storage, Data and Database Operations

1

Explain

Object, block, file and database services solve different consistency, latency and sharing needs.

2

Distinguish

IOPS, throughput, connections, partitions and caching require different evidence.

3

Apply

Apply K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.

4

Evidence

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10

TOPIC 10 · LO3 · LO4 · LO5 · K2 · K4 · A2 · A4 · A7

Performance, Cost and Integrated Troubleshooting

Bottleneck analysis · Cost Explorer · Trusted Advisor · runbooks · game days · capstone

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Key Ideas — Performance, Cost and Integrated Troubleshooting

1

Optimise after measurement

Right-size and tune using utilisation, saturation, latency and business demand.

2

Cost anomalies are operational signals

Unexpected spend can reveal leakage, misconfiguration, attack or failed cleanup.

3

Troubleshoot hypotheses

A runbook narrows evidence systematically instead of changing multiple variables.

4

Service quality is verified

SLOs, recovery tests and post-incident actions prove operational capability.

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Evidence-Led Troubleshooting



A4 STANDARD

A hypothesis is not a cause until evidence supports it and alternatives are tested.

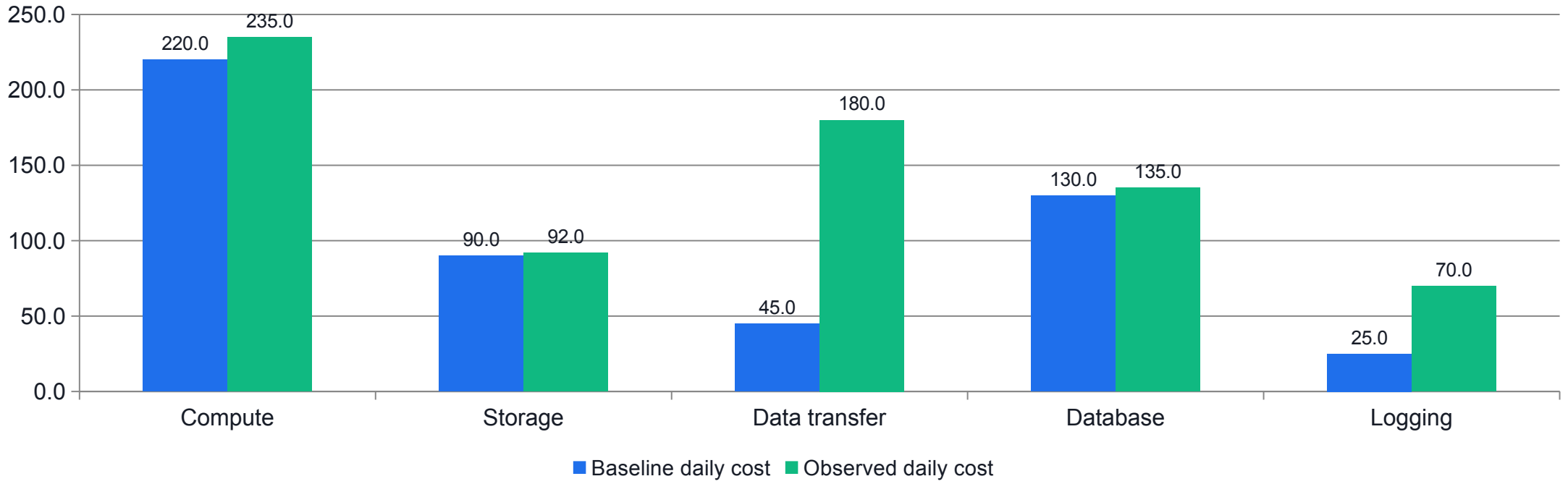
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Performance Signals

Signal	Interpretation	Example action
Utilisation	Resource use relative to capacity	Right-size only with sustained evidence
Saturation	Queued work or constrained resource	Increase capacity or remove bottleneck
Errors	Failed operations or retries	Trace dependency and change history
Latency	Time experienced by request	Inspect percentiles and request path

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Cost Anomaly Investigation



WHAT THE DATA SHOWS

The largest percentage changes are data transfer and logging; investigate traffic path, retention and recent configuration before right-sizing compute.

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Cost and Optimisation Tools

1

Cost Explorer

Trend, filter, group and forecast historical spend.

2

Cost Anomaly Detection

Surface spend patterns that deviate from expected behaviour.

3

Compute Optimizer

Resource recommendations from utilisation history.

4

Trusted Advisor

Checks across cost, security, resilience, performance and limits.

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Runbook vs Playbook

Artifact	Purpose	Contents
Runbook	Repeatable technical procedure	Inputs, steps, verification and rollback
Playbook	Coordinated response to a scenario	Roles, decisions, communication and runbooks
Game day	Exercise assumptions safely	Scenario, injects, observations and improvements
Post-incident review	Prevent recurrence	Timeline, causes, actions and verification

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Service Quality Review

Reliability

- SLO and error budget
- Recovery test results

Security

- Findings and exceptions
- Control effectiveness

Efficiency

- Capacity and cost
- Automation and toil reduction

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Capstone Standard

Restore service, explain the evidence, and improve the system.

A complete CloudOps response connects detection, diagnosis, controlled recovery, validation, communication and recurrence prevention.

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TOPIC 10 · CONCEPT-LED LAB BRIEF

Capstone — Resolve a Multi-Service CloudOps Incident

LAB 12

Apply A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting service-quality improvements.

Correlate synthetic evidence across monitoring, change, network, backup and security records; produce and execute a controlled recovery plan.

YOU'LL BUILD

A timeline, evidence matrix, prioritised hypotheses, recovery runbook, validation record and post-incident action register.

TOOLCHAIN

CloudWatch · CloudTrail · VPC Flow Logs · ELB · AWS Backup · Security Hub · Systems Manager

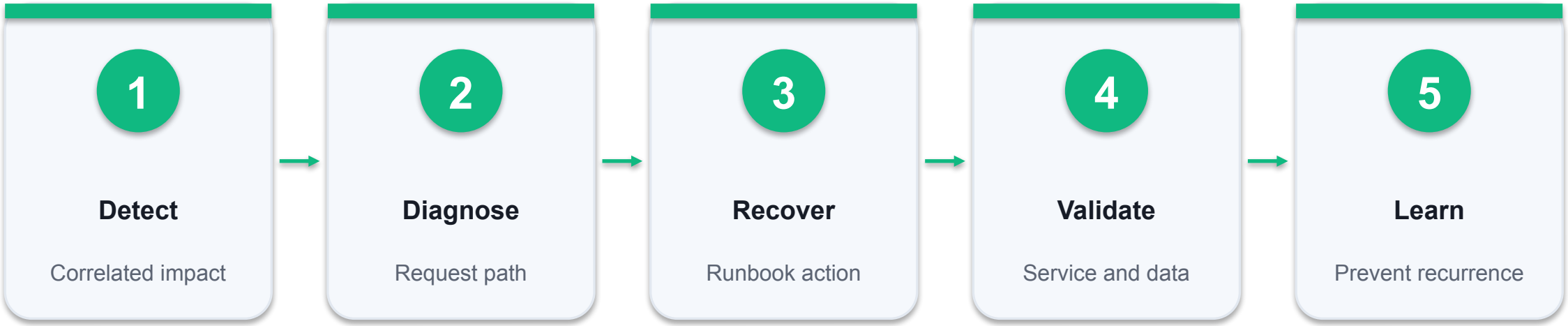
DONE WHEN

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and validation emits LAB 12 PASS.

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LAB 12 · CONCEPTUAL WORKFLOW

Lab 12 Evidence Model



PROCEDURE BOUNDARY

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LAB 12 · ACCEPTANCE EVIDENCE

Capstone — Resolve a Multi-Service CloudOps Incident

✅ Expected result

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Recap — Performance, Cost and Integrated Troubleshooting

1

Explain

Right-size and tune using utilisation, saturation, latency and business demand.

2

Distinguish

Unexpected spend can reveal leakage, misconfiguration, attack or failed cleanup.

3

Apply

Apply A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting service-quality improvements.

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END

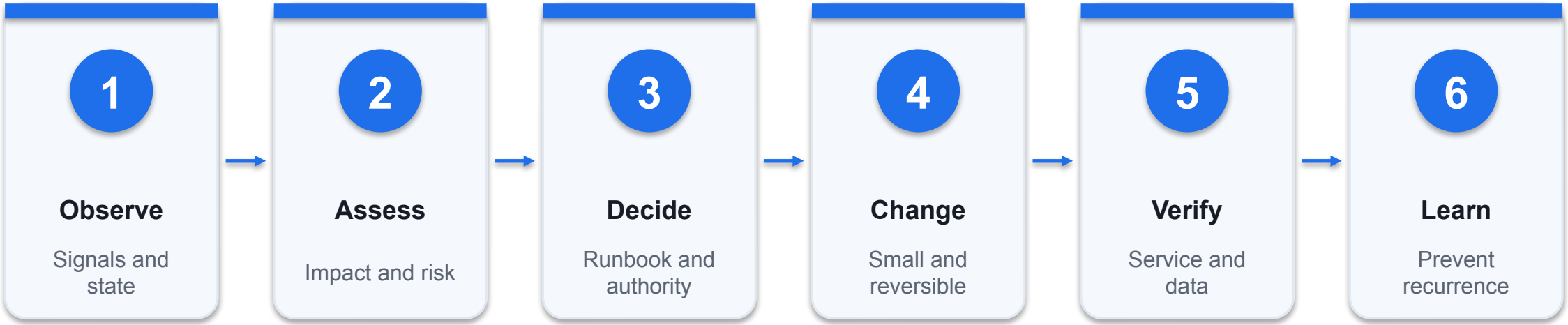
WRAP-UP

Synthesis, Assessment & Next Steps

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LO1 → LO5

The Connected CloudOps Control Loop



CAPSTONE

Lab 12 and the Case Study require one traceable operational narrative across the complete loop.

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LEARNING OUTCOMES

What You Can Now Do

1

LO1

Maintain business continuity of AWS cloud operations to ensure stable service delivery.

2

LO2

Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring.

3

LO3

Apply service quality standards to ensure security and compliance of AWS cloud operations.

4

LO4

Analyse and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.

5

LO5

Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.

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FOUR LABS PER DAY

Twelve-Lab Learning Journey

1

Day 1

Labs 01–04 · observe, triage, manage fleets and test high availability

2

Day 2

Labs 05–08 · automate change, prove recovery, enforce compliance and protect data

3

Day 3

Labs 09–12 · diagnose networks, fail over delivery, recover data and resolve a capstone incident

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CONTINUE YOUR LEARNING

Recommended Courses

1. **1**
   AWS Certified Solutions Architect Associate Training
2. **2**
   AWS Certified Developer Associate Training
3. **3**
   DevOps Engineering on AWS
4. **4**
   AWS Certified Security Specialty Training
5. **5**
   Kubernetes and Cloud Native Associate (KCNA) Training

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POST-COURSE SUPPORT

Support

1

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2

Telephone
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3

Website
www.tertiarycourses.com.sg

4

LMS-TMS
lms-tms.tertiaryinfotech.com

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READY

Assessment — Final Check

1. **WA-SAQ**
   Five open-ended questions · K1–K5 · 60 minutes.

2. **Case Study**
   Five connected tasks · A1–A7 · 60 minutes.

3. **Digital attendance**
   Complete assessment attendance and TRAQOM steps.

4. **Submission**
   Upload candidate papers through LMS-TMS; keys remain trainer-only.

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ON ASSESSMENT DAY

Assessment Flow

1. **TRAQOM**
   Course feedback
2. **Attendance**
   Assessment QR
3. **WA-SAQ**
   5 questions · 60 min
4. **Case Study**
   5 tasks · 60 min
5. **Submit**
   LMS-TMS and sign-off

APPEAL

Use the documented reassessment and appeal process if you disagree with an assessment decision.

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Digital Attendance (Mandatory)

1

AM

Scan and submit at the start of the day.

2

PM

Scan and submit after lunch.

3

Assessment

Scan before starting the final assessment.

4

Confirm

Do not close the page until submission succeeds.

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AWS CLOUD OPERATIONS

Thank You!

Operate with evidence, change safely, verify recovery and leave the system better than you found it.

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